



Central Bank
of Armenia

2024 Q3

Monetary Policy Report

Monetary Policy Program

Published September 10, 2024

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In addition to the Monetary Policy Report, the CBA concurrently publishes additional materials on its website, which the reader is encouraged to reference. These include the [MPR Chart Pack](#) (containing statistical charts and tables for global and domestic economic developments as well as the illustrative scenario projections) and the [Transparency Report](#) (providing transparency into how the Board of the CBA arrived at its decision).

DISCLAIMER The data presented in this report is current as of September 6, 2024.

01 |

Executive Summary

A. Executive Monetary Policy Statement

The Board of the CBA today reduced the Refinancing Rate to 7.50%.

At its meeting today, the Board of the Central Bank of Armenia decided to decrease the key policy rate (refinancing rate) by 0.25 percentage points, setting it at 7.50 percent. The Board agrees that a lower refinancing rate is necessary to continue to meet its price stability objective of ensuring an inflation rate of 4 percent over the medium term.

Annual CPI inflation has continued to remain below the target, registering 1.3% in August 2024. Core inflation has also remained at low levels, at 0.3% year-over-year in July.

In Q3 2024, risks of slowing economic growth globally and in the key trading partner countries of Armenia continue to persist. Global inflation continues to decline. However, sticky prices in key trading partner economies continue to remain relatively elevated. Persistent geopolitical uncertainties, as well as growing tensions in international trade relations, continue to create risks for future growth in global commodity prices and potential disruptions in global supply chains. At the same recent slowing in key trading partner countries' labor markets could point to some softening in demand conditions. In this context, key trading partner central banks would be expected to begin to gradually ease the policy rate in the near term, while still maintaining a relatively tight stance. Consequently, weak deflationary effects from the external sector on the Armenian economy continue to persist.

Economic activity in Armenia remained robust in the third quarter, continuing to be largely driven by meaningful growth in construction and trade. The latter continues to be impacted by certain short-term factors, posing significant uncertainty with respect to the sustainability of economic growth and its long-term outlook, as well as the strength of domestic demand and consumption conditions. External demand for services continues to slow relative to 2023. At the same time, risks for demand pressures stemming from fiscal policy continue to persist. Further, labor market conditions continue to cool, as reflected in the cooling pace of wage growth, and stabilizing non-traded sticky price inflation.

In order to manage possible risks stemming from conditions of high uncertainty, the Board considers multiple scenarios during its deliberations. On the one hand, the Board discussed scenarios where possible underlying developments would require a higher path for the policy rate relative to current market expectations. This includes scenarios related to uncertainty about the country risk premium and the neutral interest rate. On the other hand, the Board discussed scenarios where potential economic developments—including the risk of cooling labor market conditions driving a weak demand environment—would cause inflation to persistently remain at a low level. This would imply a more rapid and aggressive downward path for the policy rate than what is currently priced in markets in order to sustainably bring inflation back to target in the medium-term horizon.

Seeking to minimize the losses that could stem from these and other scenarios materializing, and balancing the aforementioned risks in both directions, the Board finds it appropriate to continue to gradually ease the policy stance.

The Board resolutely affirms its commitment to adopting the appropriate policy actions and strategy to ensure the price stability objective of 4% inflation in the medium term.

Approved by the Board of the Central Bank of Armenia
September 10, 2024

Governor
Martin Galstyan

Deputy Governor
Armen Nurbekyan

Board Members
Hasmik Ghahramanyan
Davit Nahapetyan
Levon Sahakyan
Narek Ghazaryan

B. Summary of Economic Conditions

Global Economy

In Q3 2024, economic activity among Armenia's main trading partners has been mixed; however, risks of weaker economic activity going forward continue to persist. Uncertainty continues to surround economic fundamentals in the US. On the one hand, despite an elevated policy rate in recent quarters, the US continues to experience strong growth, spurred by persistently robust consumption and domestic demand conditions. While inflation has slowed, sticky prices and wage growth still remain uncomfortably above their target levels, demonstrating the challenges that monetary authorities face in the "last mile." On the other hand, continued tightening of financial conditions, and the gradual depletion of excess savings, poses risks of a quick deceleration of consumption moving forward. Further, in the context of the Fed's dual mandate, risks of weakening labor market conditions imply a greater degree of readiness by the Fed to begin the loosening cycle. Looking ahead, the key uncertainty relates to the pace and magnitude of future policy easing, given financial sector vulnerabilities and still-strong consumption and demand conditions. The latter additionally raises questions about a potentially higher underlying neutral rate in the US. This would have implications on both the future stance of Fed policy and capital flows to emerging markets.

In the Eurozone, the ECB cut interest rates by 25 basis points in June, and kept the policy rate unchanged in July. However, there are only few clear signs that the inflationary environment has fully abated. On the other hand, while growth in the services sector contributed to rebounding economic activity in H1 2024, risks of economic slowdown still persist. The structurally weakened position of industry and manufacturing in key Eurozone economies remains a cause for concern. These risks, as well as concerns related to the stability of the real estate market and financial system in China, continue to remain key sources of uncertainty driving a weaker global economic outlook.

Meanwhile, the Russian economy continued to experience high growth and strong domestic demand conditions. Amid these conditions, the CBR increased policy rates by 200 basis points in July, with the potential of future rate increases. At the same time, the sustained imposition of stricter and more targeted Western sanctions continues to pose risks to the medium-term Russian economic outlook.

The inflationary environment in global commodity markets continues to remain weak, though upside risks exist. Heightened geopolitical tensions in the Middle East, which show few signs of easing, continue to threaten both higher and more volatile energy prices as well as potential disruptions to global supply chains. At the same time, food prices continue to remain below the high levels of the previous two years.

Domestic Demand Conditions

Economic growth in Armenia in Q2 2024 remained above estimates of long-run sustainable growth (approximately 5%), comprising 6.4% Y-o-Y growth. Growth was primarily driven by construction, trade, and financial services. Additionally, previous quarters' economic growth figures downward were revised downwards, owing to a reevaluation of value added within the manufacturing sector. The persistence of above-potential growth, as well as the non-widespread nature of recent growth, poses certain uncertainties regarding the relative position of aggregate demand versus aggregate supply in the economy.

The robust external demand observed since 2022 has been gradually weakening in 2024, as reflected in the stabilization of both real expenses per tourist and tourist arrivals.

In the wake of weakening global demand for IT services in recent months, production and exports in this sector have continued to decline, while some modest outflows of foreign, relatively high-skilled IT workers have been observed. Nevertheless, exports of other, "traditional" product groups—e.g. brandy, cigarettes, mineral water, and so on—remain at stable levels.

Meanwhile, considerable uncertainty continues to surround current domestic demand conditions. On the one hand, strong growth in retail trade, consumer credit, domestic tourism, and other services points to strong domestic demand. On the other hand, recent economic growth has been fairly concentrated in certain sectors, remittances have continued to decline, and labor market conditions have eased somewhat. These factors, coupled with the overall weak inflationary environment for several successive quarters, could point to weaker demand conditions in the economy.

Uncertainties also surround the outlook and future trajectory of domestic demand conditions. Key uncertainties relate to the reduced debt burden (given higher incomes); the future utilization of accumulated savings in the private sector; slowing wage growth; and geopolitical and regional uncertainties. For example, depending on how and in which direction savings are spent (consumption vs. investment), this could have different implications on the relationship between aggregate demand and aggregate supply in the economy.

Additionally, risks for modest demand pressures stemming from fiscal policy continue to persist. This is driven by risks of revenue underperformance, as well as high current expenditures given the imperative for various social support programs.

Labor Market & Inflation

The labor market has continued to cool in recent months. The household survey-based unemployment rate in Q1 2024 increased to 15.5%. Considering uncertainty surrounding the level of unemployment, it is important to continue to monitor potential development scenarios and their consequences. On the one hand, the increase in the unemployment rate could reflect arguments for weakening demand conditions across the economy, given that the uptick in unemployment has been driven by increases in the number of unemployed rather than increases in labor supply. On the other hand, the total number of registered employees, per State Revenue Committee data, continued to increase. This could suggest that labor conditions are tighter than what the household survey suggests. Alternatively, the growth in registered employees could reflect gradual, structural declines in the shadow labor market in favor of formal employment.

Slowing wage growth in recent months could support arguments of cooling labor market conditions and weakening inflationary pressures stemming from the labor market. Wage growth continued to decline in July, standing at 4.1% Y-o-Y. Moreover, wage growth has been somewhat concentrated in certain sectors, while other sectors of the economy have seen slower or flat wage growth.

Looking ahead, on the one hand, there is potential for labor force participation and supply to increase at a more meaningful pace. This could be supported by the possible weakening seasonal emigration of Armenian workers as well as a more rapid integration of forcibly displaced Nagorno-Karabakh Armenians into the labor market. Such a scenario would moderate inflationary pressures from the labor market in the medium to long term and contribute positively to potential GDP. On the other hand, the number of foreign citizens employed in Armenia has declined somewhat in recent months. If this trend were to accelerate, this could have negative impacts on both labor supply and potential GDP.

The weak inflationary environment has persisted in Q2 2024, primarily driven by moderate deflationary pressures coming from the external sector and somewhat weaker aggregate demand conditions. In this context, overall CPI inflation has remained below target since April 2023, standing at 1.3% Y-o-Y in August 2024. Non-Traded Sticky Price Inflation, which captures domestically driven demand dynamics, continued to stabilize, standing at 2.4% Y-o-Y in July 2024. At the same time, weak inflationary pressures in the services sector which are highly exposed to external demand continue to reflect trends of slowing demand. Nevertheless, high uncertainty still surrounds household inflation expectations.

In the context of the latent risks and uncertainties in the current period, the CBA builds and discusses various scenarios, summarized in the Taxonomy of Scenarios, with the aim of managing possible risks and assessing sources of uncertainty. At the same time, the MPR includes a deeper dive into two illustrative scenarios, which reflect illustrative future paths of the economy that would require either a higher path for the policy rate (Case A) or a lower path of the policy rate (Case B) relative to current market expectations. These illustrative scenarios do not represent a most-likely future, assign weight or probability to outcomes, or include all possible risks and uncertainties.

Monetary Policy

Market expectations of the CBA policy rate path have adjusted very slightly downward since the latest decisions, and continue to reflect expectations of a gradual reduction in the policy rate over the next eight decisions. Since the CBA's latest policy rate decisions, the short part of the yield curve has shifted slightly downward, while the long part of the curve remained stable. At the same time, per CBA estimates, it is likely that, besides market expectations about the future policy rate, the yield curve also reflects certain other risk factors.

The country risk premium continues to remain at relatively low levels since the beginning of the year, with some upward revisions over the last weeks, while the perceived riskiness of emerging countries overall continues to decline. This could reflect both country-specific and regional uncertainties, including tensions on the state border, regional geopolitical developments, and other factors. In this context, an upward reappraisal of the country risk premium could push assessments of the neutral rate upwards and pose inflationary risks as the stance of monetary policy stays expansionary. At the same time, macroeconomic stability in Armenia and high economic growth can serve as important preconditions for a potential reassessment of the country risk profile.

Considering the persistence of numerous types of uncertainty, the CBA builds and evaluates several different scenarios for future economic developments in order to manage possible risks stemming from these key areas of uncertainty. The illustrative Case A scenario presented in the MPR, where the policy rate path would need to be higher than market expectations, is motivated by the assumption that the country risk premium of Armenia sits somewhat lower than its fundamental level. Thus, risks of quick adjustment to long run sustainable levels are possible because of continuous regional tensions and uncertainties regarding geopolitical developments. In the event of a quick adjustment of the country risk premium, there could also emerge risks of increased volatility in financial markets and de-anchoring of inflation expectations. In this context, the CBA policy rate path would need to adopt a tighter stance relative to current market expectations in order to reflect the elevated country risk premium and neutral interest rate and guard against these adverse outcomes.

The illustrative Case B scenario presented in the MPR, where the policy rate path would need to be lower than market expectations, is motivated by the risk of a negative demand environment emerging in the domestic economy. Specifically, the increase in the unemployment rate and cooling private wage growth, as well as the weak inflationary environment, poses risks of relatively weaker longer-term demand conditions emerging. Over the medium term, these factors could pose risks of more persistent deflationary forces taking hold. In such a situation, the CBA policy rate would need to follow a sharper and more aggressive downward path than what markets presently expect, in order to stimulate demand and return inflation to the target over the medium horizon.

02 |

Monetary Policy Objectives & Framework

Statement of Long-Run Monetary Policy Objectives

Adopted effective March 12, 2024

As enshrined in “The Law of the Republic of Armenia on the Central Bank of Armenia,” the primary objectives of the Central Bank of Armenia are to ensure price stability and financial stability. The Board of the CBA is committed to using its full suite of monetary policy instruments to achieve its price stability mandate and provide a nominal anchor for the economy.

We emphasize that the primary objectives of price stability and financial stability are a means to an end. The ultimate purpose of the CBA as an accountable public-sector institution is improving the welfare of our nation. By a commitment to achieving low and stable inflation provides a credible nominal anchor for the economy and thereby anchors inflation expectations, helps to avoid inefficient boom-bust cycles, stimulates investment, supports the economic development of the country, and improves living standards over time.

The CBA develops and implements monetary policy to achieve its price stability objective. The fundamental role of monetary policy is to provide an anchor for inflation and inflation expectations. The Board believes that a four percent inflation target is consistent with its price stability mandate. The inflation target is expressed in terms of the Consumer Price Index (CPI), since it is the best measure of the cost of living and final consumption expenditures, and is the primary reference point for market participants when making planning decisions and setting contracts. Nevertheless, the overall CPI basket of goods and services includes several items that are subject to seasonal fluctuations and global economic developments, which are not directly influenced by monetary policy. We therefore consider alternative measures of inflation, such as core and non-traded sticky prices, which can provide useful information on the underlying inflationary environment.

The CBA’s primary means of adjusting the stance and path of monetary policy is through changes in the expected path of short-term interest rates. If interest rates were to be constrained by the effective lower bound, the Bank would be prepared to use its full range of policy tools.

While emphasizing the primacy of achieving the price stability and financial stability mandates over the medium term, the CBA recognizes that it is also responsible for managing the short-run tradeoff between inflation and output. In an open market economy, periodic economic and financial shocks cause economic activity and inflation to fluctuate over time. In this context, the CBA does not target CPI inflation on a period-by-period basis and does not attempt to bring CPI inflation back to the target in the short term if it would result in undue economic and financial instability. In any economic situation, the CBA seeks to develop and implement monetary policy in a manner that brings inflation to the target in the medium-term (12 quarter) horizon, while minimizing associated social costs through a “least regrets” approach.

Effective monetary policy improves welfare by anchoring long-term inflation expectations and reducing the volatility of output and unemployment. Avoiding excessive volatility in the economy can result in substantial improvements in welfare by reducing average levels of unemployment and raising the productive capacity of the Armenian economy. The Bank emphasizes that placing weight on other near-term objectives must not be inconsistent with its primary responsibilities of achieving its four percent target in the medium term and maintaining price and financial stability.

The Board places great emphasis on the importance of communicating its policy and decision-making framework to the public in a clear and comprehensible manner. Clear and effective communication helps make policy more effective, and encourages financial markets, businesses, and the general public to better understand and manage risks and uncertainties. Moreover, this communications approach plays an important role in enhancing the transparency of monetary policy and central bank accountability, which is a key priority for the CBA as an independent public sector institution.

The Board will review and renew the Statement of Long-Run Monetary Policy Objectives during the first decision-making round of each year.

Framework for Monetary Policy as Risk Management

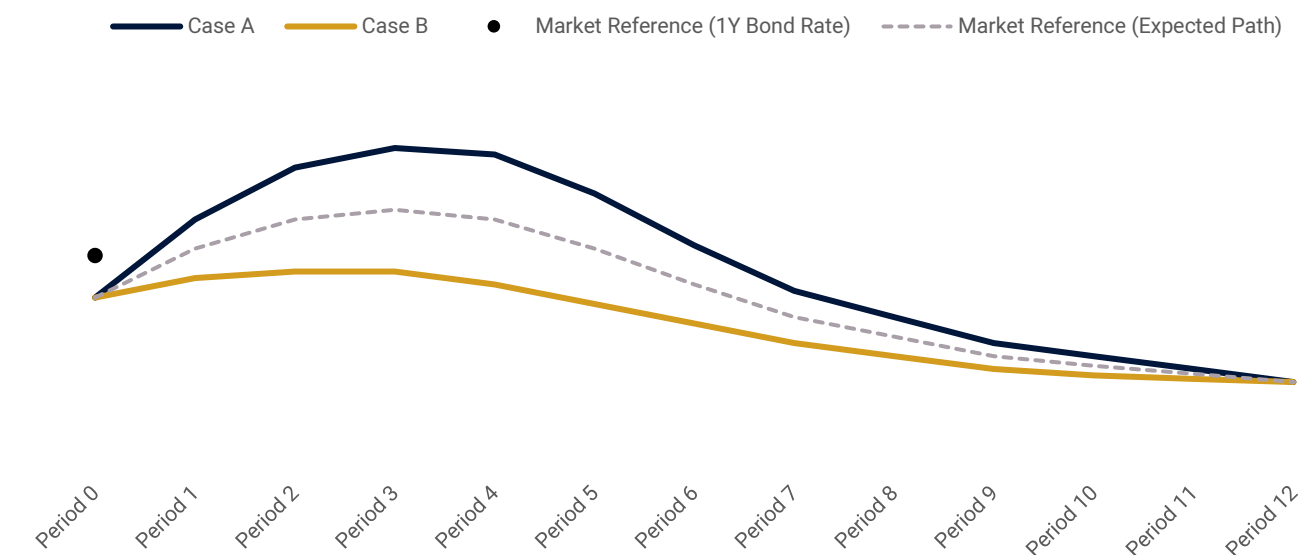
The CBA conducts monetary policy within a risk management framework to analyze and communicate the uncertainty surrounding the economic outlook more effectively. Our approach is to consider alternative scenarios for the evolution of the Armenian economy that have important implications for monetary policy. Elevated uncertainty is a reality that central banks must manage, and we do this by explicitly incorporating it into our framework and communication. We develop and analyze multiple scenarios that would imply a higher or lower path for interest rates than is currently priced in financial markets. At the same time, to steer the policy discussion and communication, we develop two or more illustrative scenarios. These scenarios should not be interpreted as pure risk scenarios, but are meant to represent plausible paths for the policy rate that could be consistent with managing monetary policy risks and concerns in an individual's economic scenario.

Market Reference is the expected path of the policy rate that is currently priced in financial markets. Ideally, if financial markets in Armenia were more developed, and there existed a derivative market for the policy rate, then we could construct an expected path based solely on market information. We therefore use the 1-year government bond yield, which should reflect the expected average policy rate over the next 12 months, plus some compensation for term premium. In addition, we conduct high-frequency surveys of market participants before, during, and following each decision-making cycle as another reference point for expectations of the policy rate.

Case A reflects a scenario that incorporates economic and financial developments that would require a higher interest rate path than what is currently priced in financial markets that is consistent with guiding the economy back to its long-run equilibrium and to achieve price stability objective.

Case B reflects a scenario that incorporates economic and financial developments that would require a lower interest rate path than what is currently priced in financial markets that is consistent with guiding the economy back to its long-run equilibrium and to achieve price stability objective.

Figure 2.1: Alternative Paths for the Policy Rate



Source: CBA

Note: For Illustrative Purposes Only

To learn more about the Central Bank of Armenia's approach to monetary policy, refer to the draft book, [Prudent Risk Management Approach to Monetary Policy: Theory and Practice of FPAS Mark II at the Central Bank of Armenia](#), edited by Douglas Laxton, Martin Galstyan, and Vahe Avagyan (2024).

Report Structure

The MPR begins with the Executive Summary, which provides the Statement by the Board of the CBA as well as a high-level assessment of economic conditions and underlying forces. The Executive Summary is also issued as a standalone document, known as the Executive Monetary Policy Statement.

Section III presents in greater detail the CBA's implementation of this policymaking framework. It begins with an assessment of the first two ingredients essential for economic analysis: where is the economy now, and what are the underlying forces driving the economy. This section also communicates how the CBA is managing current risks through a framework of "least regrets." This includes a taxonomy of different types of scenarios that may be relevant in the current policy round, reflecting a summary of the universe of major issues the Board and Staff of the CBA, as well as external stakeholders, are concerned about. Against this context, the MPR presents two or more illustrative case scenarios representing what the CBA will do with its policy instruments in order to reach its objectives.

The statistical appendix is published as a separate file, known as the MPR Chart Pack, which contains charts and tables for relevant economic variables, as well as key projections from the illustrative case scenarios.

Importantly, the MPR of the CBA is published on the same day as the policy decision is made by the Board of the CBA. This is intended to provide the public with sufficient detail and narratives that describe the framework and analysis that led to the decision in a near-real-time manner. Following the decision and press conference, a separate Transparency Report is released, containing more detailed information about how the Board came to the specific decision during the Board meeting.

Disclaimer

The information presented in this report is based on actual data available by September 6, 2024 as well as the results of the surveys conducted by the CBA and judgments made pursuant to the information on future developments of the macroeconomic environment.

The Monetary Policy Report serves as the chief monetary policy communications vehicle for the Central Bank of Armenia, and is published on a quarterly basis. The MPR has been designed to be a user-friendly document that covers key policy-relevant issues but remains accessible to technical and non-technical audiences alike. The report is structured to follow a narrative format that helps illustrate the key issues, risks, and uncertainties that drove the Board's thinking in the current decision-making round and how it arrived at its policy decision. Not all data, analysis, or issues can be reasonably included in the MPR in every quarter without overwhelming the reader or overshadowing key messages. For this reason, the MPR is supplemented with a Chart Pack that provides a complete and comprehensive overview of the current and underlying economic environment.

A decision made by the CBA Board today will affect inflation with a lag. The illustrative scenarios are chosen and designed by our Monetary Policy Department's economists to describe plausible paths of the economy under different risks. Importantly, the selected illustrative scenarios are two of many possible scenarios that the Board considers when making its decisions, which are represented in the Taxonomy of Scenarios in Section III. During each policymaking cycle, the Board is engaged in dynamic discussions and information-sharing with the Staff related to understanding the current position of the economy, its underlying drivers, and key sources of policy-relevant risk and uncertainty. These discussions, including the process through which the Taxonomy of Scenarios is developed and the illustrative scenarios are selected and quantified, are an important input for Board deliberations. The Taxonomy of Scenarios and the selected illustrative scenarios are produced four times a year and are a key, but not the only, input to the CBA Board's monetary policy decisions.

All monetary policy reports which have been published to date are available on the CBA's website (www.cba.am), which also contains all press releases, working papers, and other monetary policy-related publications.



Central Bank
of Armenia

03 |

Monetary Policy
Outlook

A. Global Economic Developments

United States Economy

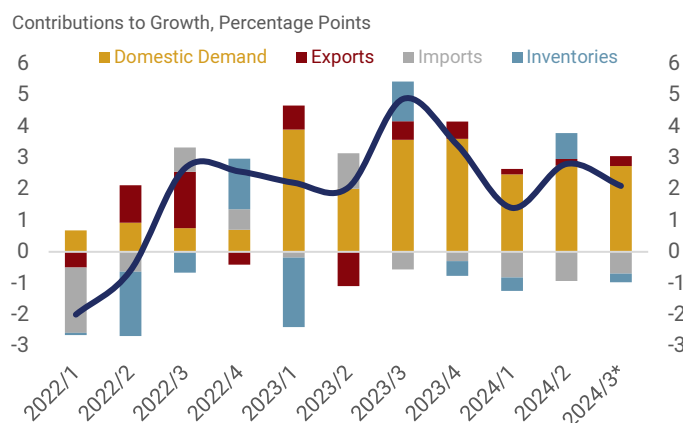
Following relatively modest output growth in the US in Q1 2024, growth in Q2 rebounded to 2.8%, well above most estimates of potential. Consumption, and in particular services consumption, continues to remain the primary driver of this growth. Considering still-elevated wage growth, high household net worth, and a somewhat expansionary fiscal stance, the preconditions exist for consumption to remain robust moving forward. These factors could drive above-potential growth in the near term, despite elevated monetary conditions. On the other hand, recent developments may give credence to counterarguments of weakening demand conditions. Delinquency rates on credit card and auto loans—which are interest rate-sensitive segments—continue to rise, while consumer confidence has been declining. This suggests that tight financial conditions may be having a restrictive effect on some segments of the economy. Additionally, the possible depletion of excess savings accumulated since 2020 could suggest less room for a further acceleration in consumption.

Inflation has continued to decline considerably in recent months. At the same time, while the pace of month-on-month growth in sticky prices has slowed, they still remain elevated. The longer sticky price inflation remains high, the greater the risks become of inflation expectations remaining de-anchored from the target and influencing the price-setting behavior of firms and workers. In this context, wage growth continues to remain elevated at levels that are not wholly consistent with a sustainable path back to the inflation target. Alternatively, recent developments in the labor market could suggest greater cooling, significant uncertainty notwithstanding. Hiring has continued to slow, while the number of job openings per unemployed person and the job quits rate have been falling. On the other hand, while the unemployment rate (at 4.2% in August) now stands above its pre-Covid levels, recent increases have mostly been driven by growth in labor supply rather than layoffs, suggesting that true rebalancing in the labor market has not yet taken place.

Markets in the US have become significantly more optimistic of future rate cuts in the near term (Figure 3.A.3), driven by rising concerns that the Fed has fallen behind the curve in cutting rates. At Jackson Hole, Chairman Powell suggested greater confidence that inflation is on a sustainable path back to the target. This, as well as growing downside risks of labor conditions weakening in the context of the Fed's dual mandate, would imply a greater readiness by the Fed to begin the loosening cycle. However, uncertainty about certain economic fundamentals, concerns about financial sector vulnerabilities, and still-strong consumption and demand conditions could temper the magnitude of near-term rate cuts relative to market expectations. Additionally, uncertainty about the underlying neutral rate (refer to the CBA's Q2 2024 MPR) raises important questions about where policy rates may land in the long-term.

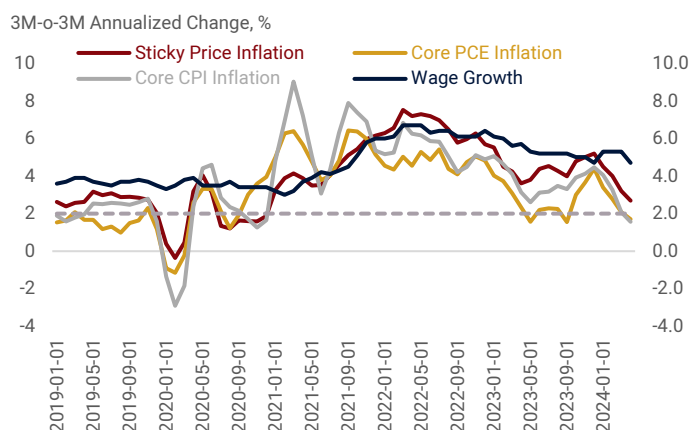
Robust Consumption Continues to Drive Above-Potential Growth in US, but Outlook Remains Uncertain

Figure 3.A.1: Contributions to US Real GDP Growth



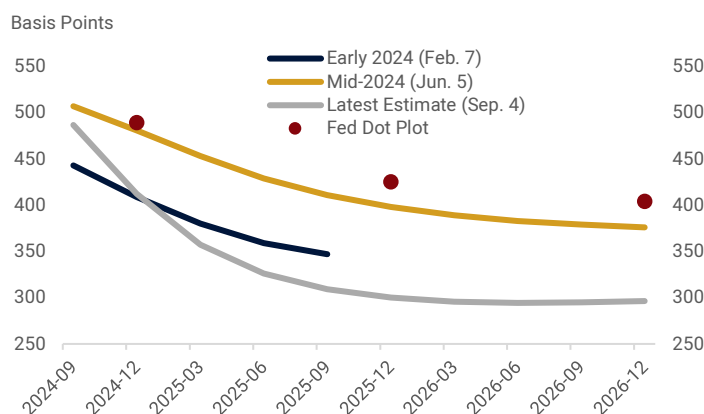
While Inflation has Declined in Recent Months, the "Last Mile" Poses Serious Challenges for the Fed

Figure 3.A.2: Underlying Inflation Measures, US



Markets have become Significantly More Optimistic of Imminent Fed Rate Cuts

Figure 3.A.3: Market Expectations of FED Funds Rate Path



Eurozone Economy

The Eurozone economy has exhibited signs of modest recovery in economic growth in H1 2024, though the outlook continues to remain highly uncertain. Seasonally adjusted GDP grew by 0.6% Y-o-Y in Q2 2024, and 0.2% Q-o-Q (Figure 3.A.4). Household final consumption (up 0.5% Y-o-Y), and in particular services consumption, continue to remain important drivers of growth. The persistence of demand-driven growth suggests that imbalances between aggregate demand and supply may still exist.

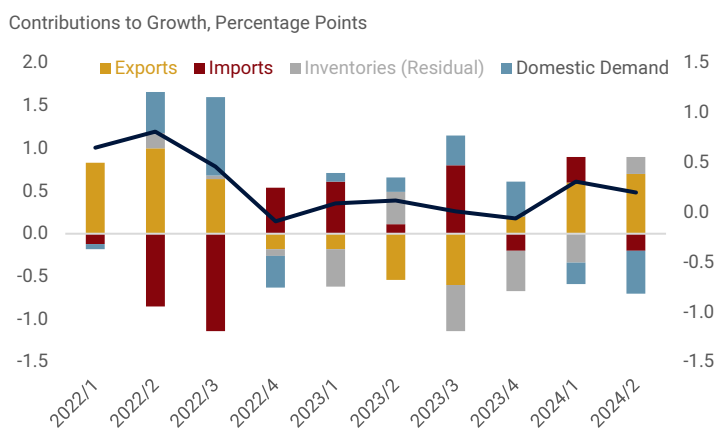
The services sector remains the sole source of economic growth in the Eurozone. The pace of service sector expansion increased in August (high-frequency PMI indicators for services are at a three-month high; Figure 3.A.5). In this context, the combination of increasing real disposable income levels and still-high savings could fuel growth in consumption and drive further services-driven expansion moving forward. Meanwhile, manufacturing and industry continue to remain in a structurally weakened position, especially in key industrial economies such as Germany and France. Manufacturing continues to be hampered by tight monetary conditions, sluggish recovery from the energy crisis, and the risk of loss of competitiveness in international markets in recent years. Developments in recent months—including falling new orders and accelerating contractions in output and employment—suggest that these structural uncertainties could continue to pose downside risk to the medium-term Eurozone growth outlook.

The inflationary environment in the Eurozone continues to stabilize, with inflation growing at 2.2% Y-o-Y in August 2024 (Figure 3.A.6). However, underlying inflationary measures, particularly services inflation—which is most sensitive to domestic demand conditions—remain persistently elevated at around 4% for several months, potentially driven by labor market pressures. The unemployment rate in the Eurozone declined to 6.4% in July, remaining close to historic lows. This suggests that labor market conditions still remain tight, and many firms continue to report labor shortages.

The combination of persistent underlying inflationary pressures, along with structural concerns about the economic outlook, underscores the difficulties the ECB faces in the “last mile” as it seeks to bring inflation sustainably back to target. While the ECB reduced policy rates by 25 basis points in June, it has refrained from providing explicit forward guidance, affirming that it is “not pre-committing to a particular rate path.”

A Stagflationary Environment in the Eurozone Could Test the Resolve of the ECB to Bring Inflation on Target in a Timely Manner

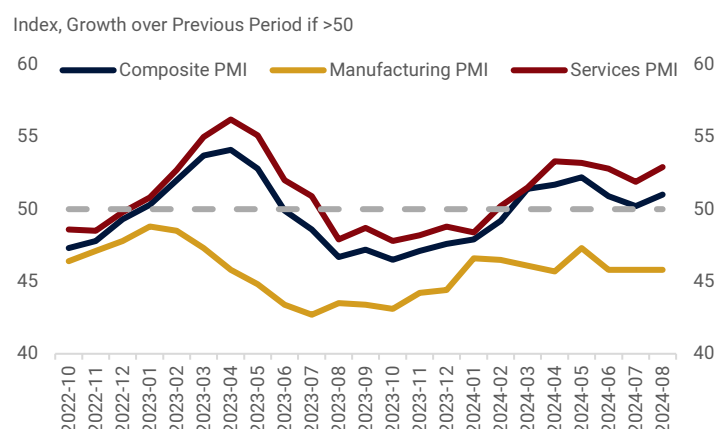
Figure 3.A.4: Contributions to Euro Area Real GDP Growth



Source: Eurostat;

Eurozone Growth Continues to be Driven by Services, with Manufacturing in a Structurally Weakened Position

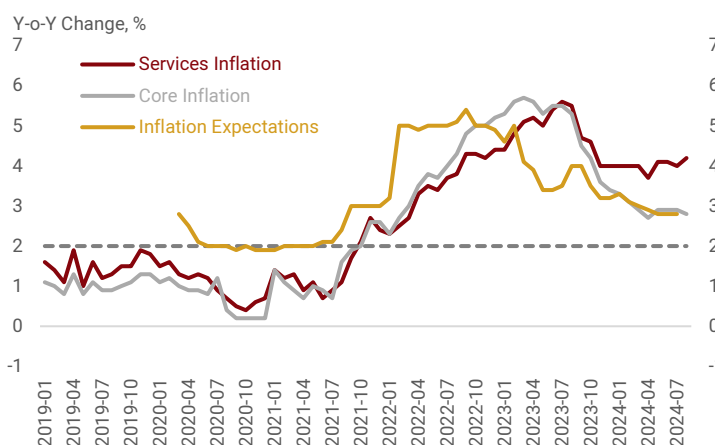
Figure 3.A.5: Composite Purchasing Managers Index (PMI), Euro Area



Source: S&P Global & Hamburg Commercial Bank

Underlying Inflation Measures Differ Substantially from Each Other, Presenting Very Different Views about the Economy

Figure 3.A.6: Underlying Inflation Measures, Euro Area



Source: Eurostat

Russian Economy

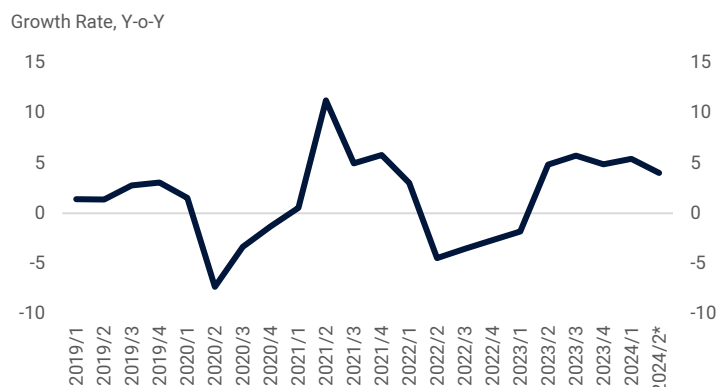
While the Russian economy grew at a somewhat slower pace in Q2 (4.0% Y-o-Y) relative to previous quarters, the Russian economy continues to remain in an overheated position (Figure 3.A.7). Several sectors continue to experience robust growth—specifically wholesale and retail trade as well as manufacturing (especially defense industries). The main growth driver remains domestic demand, amid a significant increase in household incomes in recent quarters. These resilient demand conditions are underpinned by exceedingly hot labor market conditions, with the unemployment rate reaching record lows (2.4% in July) and real wage growth consequently rising rapidly. In this context, domestic demand continues to outstrip producers' capacity to meet demand for goods and services. The expansion of production capacities is further exacerbated by labor shortages, driven by intensified competition for recruits between the military and businesses.

Given these excess demand conditions, inflation remains stubbornly high, rising to 9.1% Y-o-Y in July, propelled by increases in housing and utility costs. Inflation expectations continue to surge as well, with households anticipating an average inflation rate of approximately 12.9% over the next 12 months. Amid these conditions, the Bank of Russia increased policy rates by 200 basis points in July, with the central bank suggesting that additional rate increases may be imminent. However, the ability of higher interest rates to curb excess demand conditions is hampered by a highly expansionary fiscal policy stance. Because government expenditures are less responsive to higher borrowing costs, there are risks that a tighter policy stance may not effectively curb inflationary pressures. On the other hand, the phasing out of state-subsidized mortgage programs in June (which shielded consumers from high rates) could pose downside risks to the construction and banking sectors and the overall growth outlook.

In addition to the risks noted above, several medium-term risks pose challenges for the economic outlook. Efforts by Western countries to more strictly enforce existing sanctions (e.g. oil price cap) and to implement new, more targeted sanctions (affecting both oil and gas and other strategic industries, as well as third countries that aid in circumventing sanctions) throughout 2024 pose several risks. First, increasingly targeted sanctions could threaten budget revenues from both oil/gas and other sectors, the latter of which has come to play an increasingly important role. Moreover, any lost revenues could be supplemented by higher domestic tax collection, which could drag down domestic consumption and demand. Second, amid limited access to international markets, more targeted sanctions threaten Russia's ability to efficiently substitute imports of certain strategic products with domestically-produced goods, including investment goods for both civilian and military use, which could have negative impacts on medium- and long-term potential.

Domestic Demand Continues to Outstrip Capacities to Expand Production of Goods and Services

Figure 3.A.7: Russia Real GDP Growth

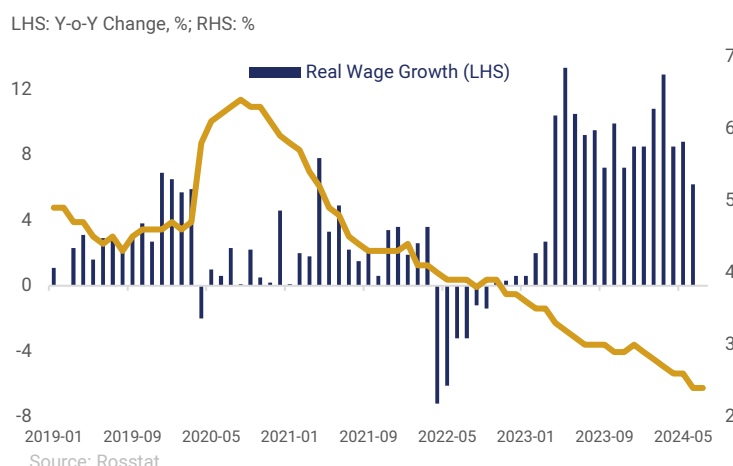


Source: Bank of Russia

Note: Data for 2024Q2 is a preliminary estimate.

A Low Unemployment Rate and High Real Wage Growth Lend Credence to Ongoing Labor Shortages that Are Constraining the Expansion of the Output of Goods and Services

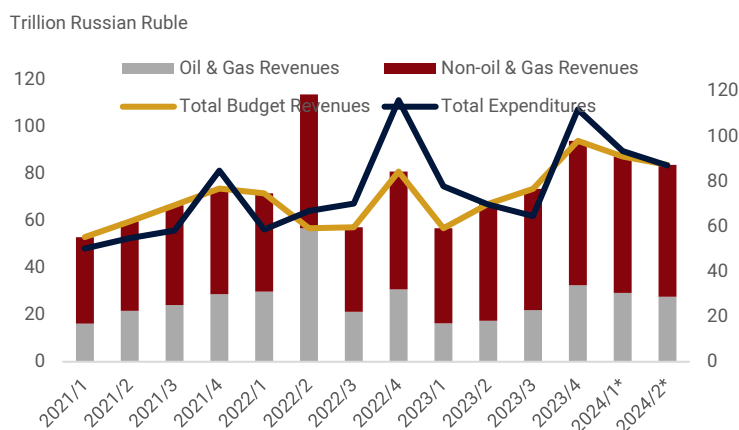
Figure 3.A.8: Unemployment Rate and Real Wage Growth, Russia



Source: Rosstat

Despite Ongoing Sanctions and Surges in Defense Spending, Russia has Broadly Avoided Deepening the Budget Deficit

Figure 3.A.9: Russian Budget Revenues and Expenditures



Note: Initial estimates for 2024 Q1 and Q2.

Source: Russian Ministry of Finance

Commodity Markets

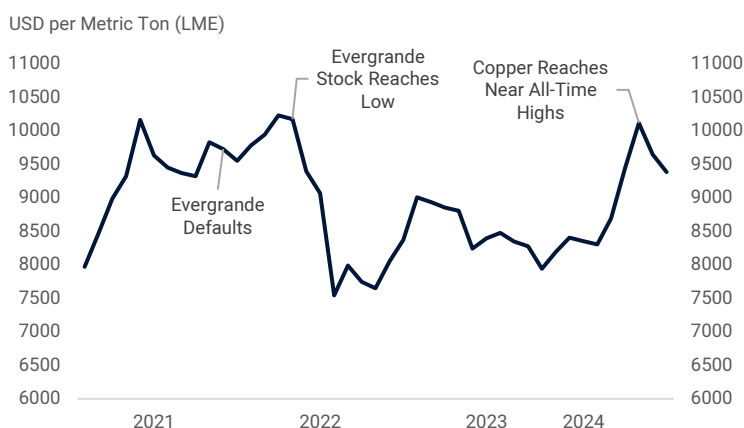
Copper: Following a steady decline in copper prices in 2023 due to slowing global demand (particularly in China, the main buyer of copper), copper prices rebounded in H1 2024, reaching near all-time highs in May of approximately 10,120 USD per metric ton. Much of this price growth stemmed from the effects of supply disruptions/constraints in early 2024. Since then, copper prices have steadily declined, standing at 9,385 USD per metric ton in July, largely driven by an uncertain outlook for global demand, including continued weak economic activity in China. The outlook for copper remains characterized by significant volatility, driven by both macroeconomic and industry/country specific dynamics. On the one hand, the global transition to green technologies continues to drive long-term growth in demand. On the other hand, risks of increased regulations and trade barriers, as well as uncertainty about global demand conditions, are an important source of risk and uncertainty moving forward.

Oil: Global oil prices have exhibited volatility in recent months, driven by uncertainty about the global macro outlook, geopolitical developments, and supply. Continued weak demand from China, as well as uncertainty about the trajectory of economic activity in advanced economies, pose downside risks to oil demand and prices. Despite these developments, OPEC+ has yet to commit to a timeline for unwinding its supply cuts, though supply increases by non-OPEC countries support sustained growth in global oil supply over the medium term. On the other hand, uncertainty around the geopolitical outlook in the Middle East, where one quarter of global oil is produced, still poses serious upside risk, particularly given the potential for regional escalations.

Food: The FAO Food Price Index (FFPI) stood at 120.7 points in August 2024, down slightly versus July. This decrease was driven primarily by declines in sugar, meat, and cereals prices. Declines in sugar and cereals prices were primarily driven by favorable weather conditions and an improved production outlook in key producer countries. These decreases offset slight increases in vegetable oil and dairy prices. Food prices generally remain well below the levels of the prior two years (FFPI down 1.1% Y-o-Y in August, and down 24.7% versus its peak in March 2022). However, sustained risks of disruptions in supply chains, including those related to geopolitical tensions in the Middle East, continue to drive upside risk, which could have implications on imported food prices in Armenia.

Following Spike in May, Copper Prices have Declined, Reflecting Uncertainty about Global Macro Fundamentals

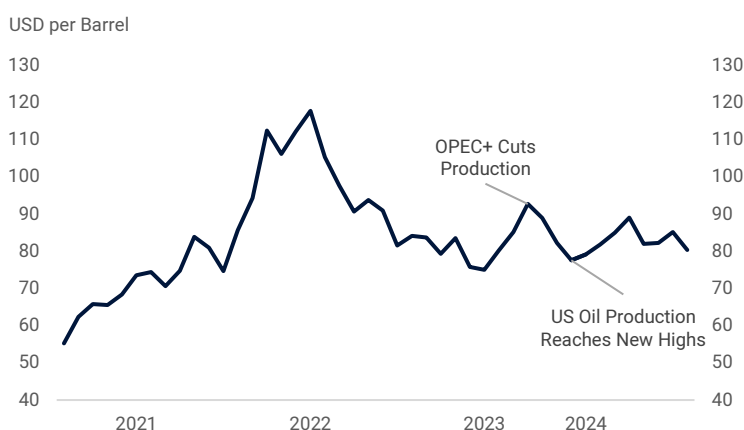
Figure 3.A.10: Copper Prices



Source: FRED

Amid Sustained Geopolitical Tensions, Questions about Global Demand, and Potential Growth in Supply, Considerable Uncertainty Remains for the Global Oil Outlook

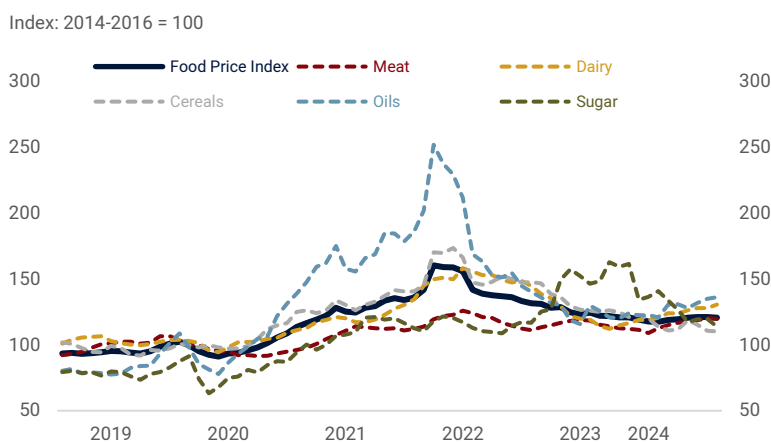
Figure 3.A.11: Brent Crude Oil Prices



Source: FRED

Global Food Prices Have Significantly Moderated as Supply/ Demand Normalized Post-Covid

Figure 3.A.12: FAO Food Price Index



Source: FAO

Table 3.A.1: Third-Party Projections for Commodity Prices

Source/ Topic	Oil (Brent, Dollars per Barrel)			Copper (LME \$/t)			World Bank Food Price Forecasts (Ann. Growth)		
	EIA	WB	Consensus	WB	REQ	Consensus	Oils and Meals	Grains	Other Food
	Aug 24	Apr 24	Sep 24	Apr 24	Jun 24	Sep 24	Apr 24	Apr 24	Apr 24
2024	83	84	83	8900	9510	9420	-7.3	-11.3	2.1
2025	80	79	80	8800	9860	9952	-4.9	-3.8	-2.8

Source: US Energy Information Administration; World Bank; Australia Resources and Energy Quarterly (REQ); Bloomberg

Table 3.A.2. Third-Party Projections for GDP Growth and Inflation in Key Trading Partner Countries

Source	United States					Euro Area						Russia				
	IMF*	WB	OECD	Fed	Consensus	IMF*	WB	OECD	EC	ECB	Consensus	IMF*	WB	OECD	CBR	Consensus
	Jul 24	Jun 24	May 24	Jun 24	Sep 24	Jul 24	Jun 24	May 24	May 24	Jun 24	Sep 24	Jul 24	Jun 24	May 24	Jul 24	Sep 24
GDP Growth (Annual Rate, %)																
2023			2.5 (Actual)					0.5 (Actual)					3.6 (Actual)			
2024	2.6	2.5	2.6	2.1	2.5	0.9	0.7	0.7	0.8	0.9	0.7	3.2	2.9	2.6	3.8	3.3
2025	1.9	1.8	1.8	2.0	1.7	1.5	1.4	1.5	1.4	1.4	1.3	1.5	1.4	1.0	1.0	1.6
2026	2.0	1.8	-	2.0	2.0	1.4	1.3	-	-	1.6	1.4	1.2	1.1	-	1.5	1.3
Inflation (Annual Rate, %)																
2023			3.4 (Actual)					2.9 (Actual)					7.4 (Actual)			
2024	2.9	-	2.6	2.6	2.9	2.4	-	2.6	2.5	2.5	2.4	6.9	-	7.2	6.8	7.5
2025	2.0	-	2.1	2.3	2.3	2.1	-	2.1	2.1	2.2	2.1	4.5	-	5.3	4.3	5.2
2026	2.1	-	-	2.0	2.3	1.9	-	-	-	1.9	2.0	4.0	-	-	4.0	4.2

Note: IMF inflation forecasts per April WEO.

Source: IMF; World Bank (WB); OECD; Fed; ECB; CBR; European Commission (EC); Bloomberg

B. Domestic Demand Conditions

Gross Domestic Product

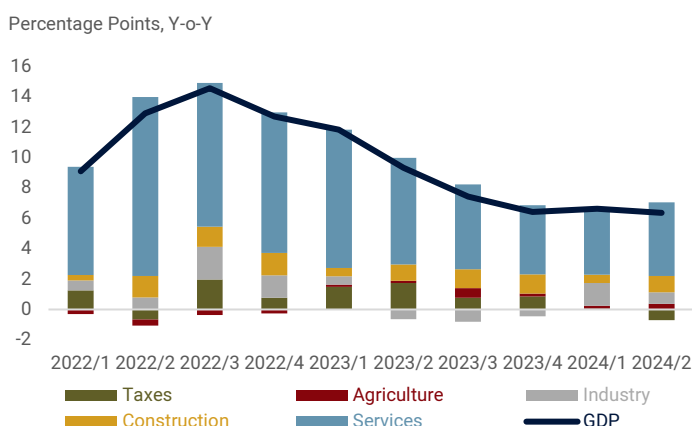
Economic growth in Q2 2024 stands at 6.4% Y-o-Y, and is largely concentrated in the trade (19.0% Y-o-Y), financial services (17.0% Y-o-Y), and construction (17.9% Y-o-Y) sectors. GDP growth was significantly revised downwards in previous quarters which was mostly due to the revision of manufacturing value added. Growth continued recovering in the agriculture sector, with a Y-o-Y growth rate of 5.2%, and is in positive territory in the manufacturing sector for the 1st half of 2024, despite the downward revisions, after contracting in the previous couple of quarters. The persistent above potential economic growth, albeit concentrated in some sectors and not broad based, poses key uncertainties and risks related to assessing the relative state of aggregate demand versus aggregate supply. After several years of positive growth in the information and communication services sector, contraction was observed in the first half of 2024. Additionally, a certain level of outflow of foreign and relatively more skilled labor in the industry is noticeable. This downward adjustment can have medium-term implications given the sector's crucial role in driving and expanding the export capacities and potential output of the country. However, there are no clear signs as of yet of a widespread contraction, since the recent decline in Armenia, which is still lower compared to the global tech slowdown of the past two years, may be the organic result of AI-related developments and more global trends. Further, trade has emerged as the main driver of services growth, suggesting that either demand-driven factors are outpacing supply, or that large shares of growth are concentrated in re-exporting activities.

The robust external demand observed since 2022 has gradually weakened by H1 2024, as figures reflecting the stabilization of tourist arrivals and decline of real expenses per tourist indicate (Figure 3.B.2). Considering that most of the tourists visiting Armenia are from regional countries, uncertainties regarding the growth outlook of Russia and regional geopolitical developments could pose risks of slowdown in the tourism sector, with potentially cooling effects on economic activity and slowing export growth.

In Q2 2024, the output gap is estimated to be largely closed (Figure 3.B.3), though uncertainties regarding estimates remain. On the one hand, sustained above-potential economic growth, elevated private spending, and high loan growth suggest that the economy is operating under excess demand conditions. On the other hand, the concentration of this recent growth in a few sectors, easing labor market conditions, and the weak inflationary environment could point to weak demand conditions, and even suggest a negative output gap. Moreover, the cyclically appreciated real effective exchange rate (Figure 3.B.8) has also helped stem excess demand pressures from the external sector.

Growth Continues to be Concentrated in Some Sectors Raising Questions about the State of Domestic Demand Conditions

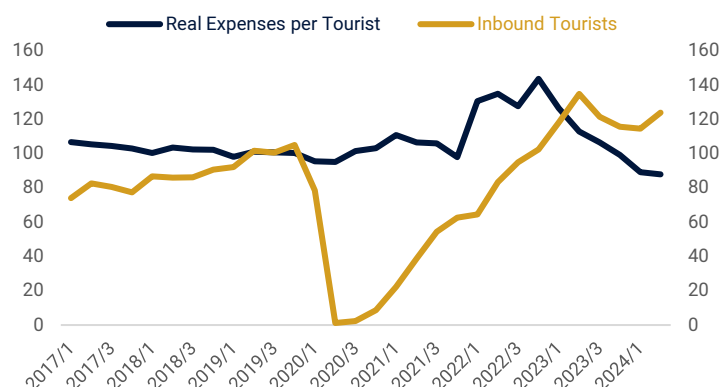
Figure 3.B.1: Contribution to Real GDP Growth by Production Approach



Source: Statistical Committee of the Republic of Armenia; CBA

Both Inbound Tourism and Real Expenses per Tourist are Normalizing

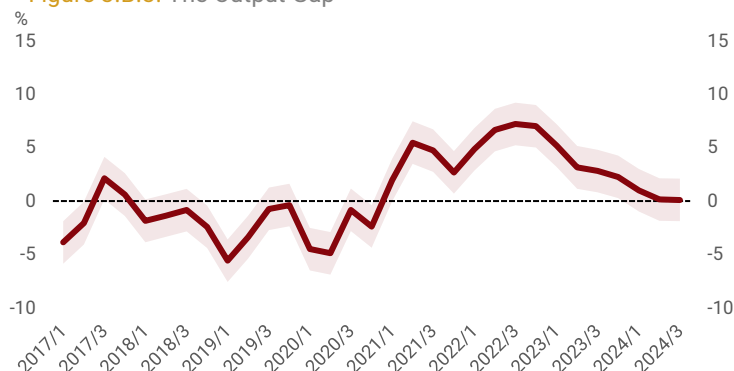
Figure 3.B.2: Real Expenses per Tourist and Inbound Tourists, Armenia
Index: 2019 = 100



Source: Statistical Committee of the Republic of Armenia; CBA

Aggregate Demand Continues to Outstrip Supply

Figure 3.B.3: The Output Gap



Source: CBA

Consumption and Investment

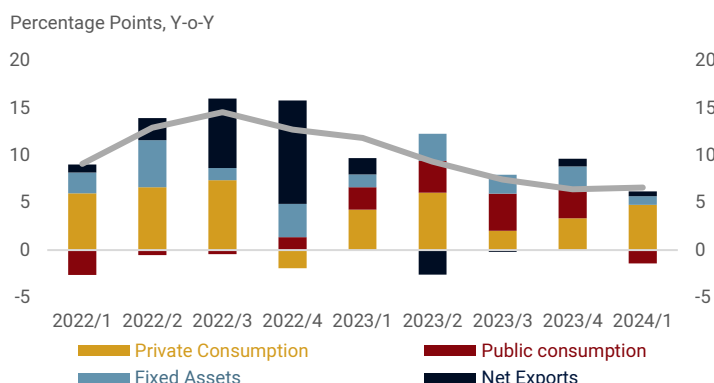
In the second quarter of 2024, growth in aggregate consumption was primarily driven by an increase in private consumption, even as government spending declined (Figure 3.B.4). Private consumption, which had been relatively subdued in relation to overall economic growth in the past couple of years, gained momentum in the 1st half of 2024, achieving a growth rate of 7.5% Y-o-Y. Aggregate investment growth, which was largely driven by growth in public investment through the end of Q1 2024, accelerated in the second quarter as both public and private investments grew. Private investment in the real estate sector continue to remain at high levels. Investment activity, more broadly, has recovered somewhat, as indicated also by increasing loans provided to businesses. In this context, the high number of construction permits issued suggests that private investment growth can continue further into the future (figure 3.B.6). Conversely, slowing imports of investment goods in the second quarter could provide signals about potential delays in investments, mainly due to ongoing regional and geopolitical uncertainties.

The main uncertainties and risks in the near term relate to the trajectory of domestic demand. On the one hand, domestic demand could grow on the back of previously accumulated savings in the private sector, a reduced debt burden, and subsequent potential for credit growth. Such a scenario could fuel further excess demand pressures (particularly if the economy is judged to be already operating at or slightly above potential), increasing the inflationary pressures. On the other hand, non-widespread economic growth, the increase in the unemployment rate, and slowdown in wage growth in the private sector brings about risks of weak demand conditions, which will be reflected in the weak inflationary environment. Moreover, a spike in regional and geopolitical uncertainty can restrain private expenditures in the form of both consumption and investments. This scenario implies aggregate demand being below long-term potential. Another important consideration relates to the impacts of the inflow of forcibly displaced people from Nagorno-Karabakh on aggregate demand. Social assistance programs to support their socioeconomic integration are expected to have some positive effect on aggregate demand in the near term. Their integration into the labor market will be a crucial factor in increasing potential GDP and in balancing demand and supply forces in the economy.

Remittances of individuals continued to decline considerably (Figure 3.B.5), due to the slowdown of economic activity in trading partner countries. At the same time, the narrowing wage gap between the construction sectors of Armenia and Russia (Figure 3.C.3), as well as the potential taxation of migrant workers' incomes earned abroad could create risks of potential slowdown in seasonal migration and remittances. Therefore, growth in labor supply and potential output is possible due to the integration of seasonal migrant workers into the Armenian labor market which would bring about deflationary risks.

Recent Economic Growth is Not Broad-Based, Posing Downside Risks to Consumption and Demand Conditions

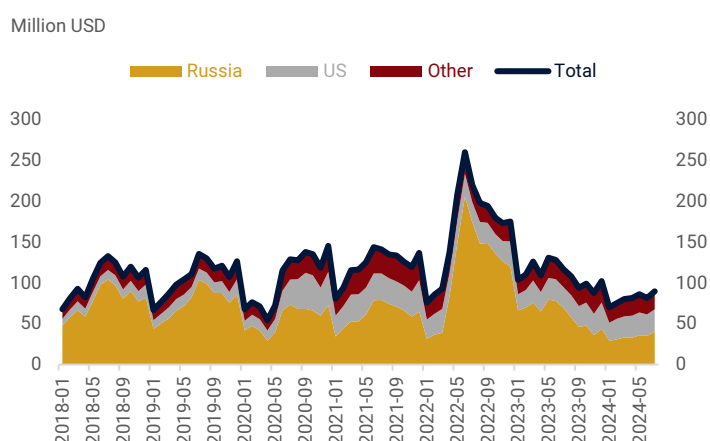
Figure 3.B.4: Contribution to Real GDP Growth by Expenditure Approach



Source: Statistical Committee of the Republic of Armenia; CBA

Remittances Continue to Decline Considerably from 2022 Peaks

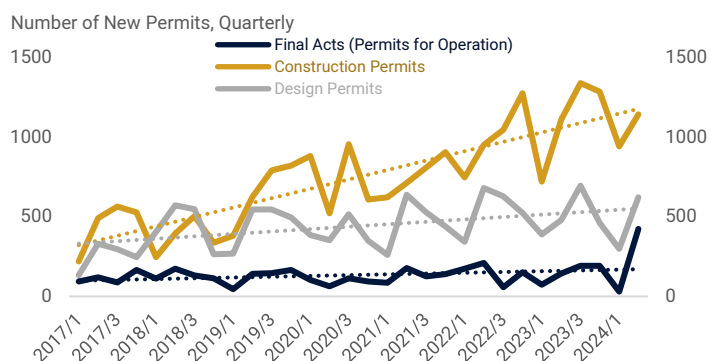
Figure 3.B.5: Inflow of Non-Commercial Transfers to Armenia by Source Country



Source: CBA

Construction Permit Growth Continues at a Record Pace, Supporting Investment Activity in the Construction Sector

Figure 3.B.6: The Number of Issued Design and Construction Permits and Completion Certificates in Yerevan



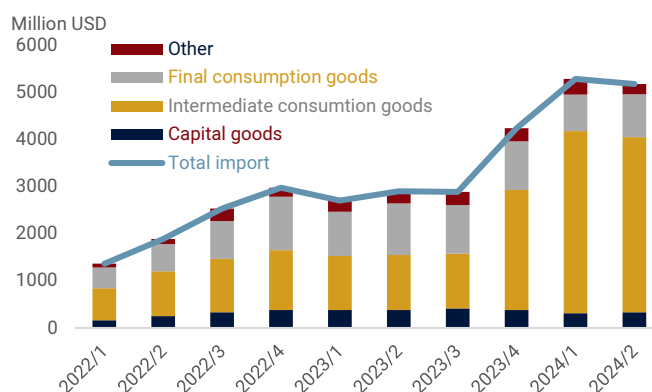
Source: Yerevan Municipality

External Trade and Demand

In Q2 2024, strong growth was sustained in exports and imports, primarily driven by increased trading activity in certain sectors. At the same time, strong exports of domestically produced goods (in line with the robust levels observed in the past two years) reflect sustained strong external demand for Armenian goods, though adjustments have been observed for some products. Despite the strong level of savings accumulated in recent years and high economic growth, demand for imported goods has somewhat weakened, though it is not broad based (for example, declining motor vehicle imports mainly reflect previous market oversupply). On the other hand, it is also possible that the strong income growth in recent years could be generating shifts in consumption preferences from goods to services. With respect to restrained demand for investment goods imports, possible reasons could include uncertainty related to regional developments, as well as the drawdown of certain investment projects. Overall, these developments further add to uncertainties about the current state and future trajectory of domestic demand conditions.

The real effective exchange rate (REER) depreciated somewhat in Q3 2024, following appreciation of the currency basket of main trading partners' nominal exchange rate (especially Russian ruble, which has appreciated during 2024 Q3 after a sharp depreciation). The key uncertainties relate to the cyclically appreciated exchange rate in the context of stabilizing external imbalances and absorbing excess demand. Elevated external demand since 2022 amid geopolitical developments in Eurasia, capital and labor inflows, high tourist inflows and export of highly productive IT services have contributed to a significant appreciation of the REER in the economy (Figure 3.B.8), while weak inflation in the domestic economy versus partner countries since 2023 has contributed to lowering REER appreciation. If REER appreciation were to prove to be cyclical, its parallel adjustment would inevitably be reflected in a sharp adjustment of the current account deficit, following the aggregate demand correction process. On the other hand, it may be that current levels of external demand reflect new opportunities that would permanently support export potential (especially of services), driving fundamental (rather than cyclical) REER appreciation. This notion is supported by growth in supply and export capacities during this period. This scenarios and policy implications are described in box 2.

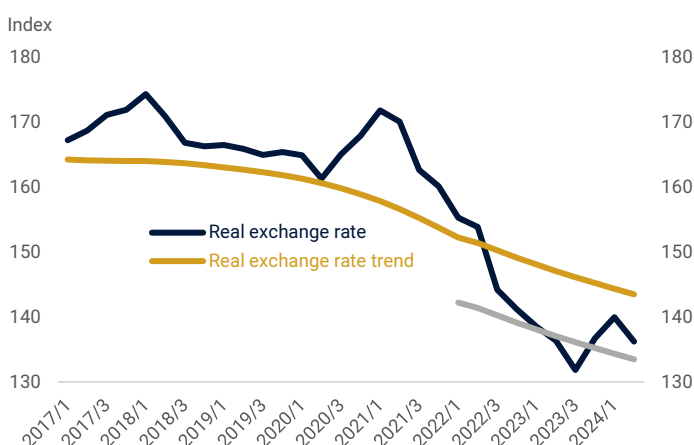
Demand for Consumption & Investment Goods Remains weak
Figure 3.B.7: Value of Imports by Armenia by Broad Economic Category (BEC)



Source: Statistical Committee of the Republic of Armenia; CBA

Uncertainty Regarding the Cyclical Position of the Real Exchange Rate Persists

Figure 3.B.8: Real Effective Exchange Rate and Trend for AMD against USD, EUR and RUB (Lower Value means Stronger Dram)



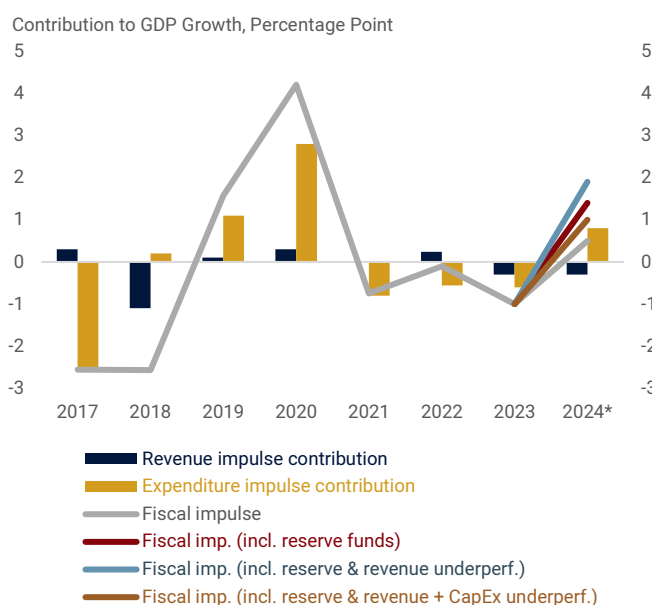
Source: CBA

Fiscal Policy

Despite the strong GDP numbers since the beginning of 2024, government tax revenues are underperformed compared to the initial plan due to recent structure of economic. Although tax revenues has increased (7.6% y/y in 2024H1) compared to the first half of the previous year, some of the increase is attributed to the methodological change of accounting for income tax refund on mortgages both in tax revenues and government expenditures. Accounting for methodological changes the growth rate of tax revenues in 2024H1 amounts to 4.3%, which is somewhat lower compared to the GDP growth at the same period. Underperformance of tax revenues creates some risks of running a higher deficit and higher public debt in 2024 and making fiscal policy more expansionary than planned. Alternatively, fiscal authorities would need to take a more conservative stance on expenditures from the perspective of balancing the budget, though most of the current expenditures are not revertible, while cutting back on capital expenditures poses some risks on medium term growth potential. Over the medium term horizon, in order to finance the budget deficit, there are certain risks of a stricter tax administration and tighter tax legislation policies, which can have an additional negative impact on aggregate demand.

Inflationary Impacts of Fiscal Policy are Relatively Muted, though Inflationary Risks Could Stem from Revenue Underperformance, Underspensing of Capital Expenditures, and/or Reserve Fund Drawdown, as well as Possible Increase in Public Debt

Figure 3.B.9: Fiscal Impulse and Scenarios for 2024



Source: Ministry of Finance; CBA calculations

Box 1. The Armenian Economy's Future Development Outlook: Illustrative Scenarios

Following the onset of the Russia-Ukraine conflict, large capital, labor, and financial inflows, as well as high demand for Armenian-produced goods and services, have laid the foundation for strong economic and income growth in Armenia. Between 2022 and 2024, real incomes have grown by 30%, while economic growth has averaged approximately 10% per year. Given these positive economic developments, the key uncertainty moving forward relates to the interpretation of recent history, which can have vastly varying implications on the future economic outlook and, consequently, on appropriate economic policy responses.

Against this background, several possible scenarios exist for Armenia's future development outlook, depending on the underlying interpretation of recent economic developments. This box presents three such illustrative scenarios (demand-driven growth; low-value productivity growth; and high productivity growth), which have meaningfully different implications for the future course of economic growth and policy responses. Importantly, the scenarios described below do *not* represent the three most likely paths for economic development, but rather, provide an opportunity to critically examine developments in the economy and implement policies of "least regrets" through a prudent risk management decision-making framework.

continued on the following page

Scenario 1. Economic “Bubble”

The primary source of economic growth since 2022 has been the emergence of very high external demand for Armenian goods and services (see Section 3.B). This high external demand environment has contributed to high income growth in selected sectors (e.g. tourism, finance, real estate) as well as an expansion of the domestic inflationary environment (e.g. services, real estate prices, and so on; Figure B.2.2). In this type of scenario, it would be essential for the economy to correct itself back to its long-term, sustainable growth path as quickly and manageably as possible. Under this interpretation, local currency appreciation would represent an “automatic” stabilizer from the perspective of managing high external demand, and promotes the economy returning back to its sustainable growth path. Importantly, such a correction would imply a rapid and meaningful slowdown in economic growth, even below potential (estimated at 4.5-5.0%), in the relatively near term. It is important to note that in this scenario, this potential slowdown in the pace of growth would be a consequence of a natural and necessary correction of the economy. Any economic policies aimed at expanding demand conditions—whether in the form of low interest rates, sectoral subsidies, higher government expenditures, and so on—in order to boost growth would only worsen the economic “bubble,” creating real risks of causing a financial/economic crisis in the future.

Figure B.2.3 presents an illustrative scenario of how different economic variables would respond to the external demand shock, assuming two possible policy responses: first, inflation targeting, and second, temporarily fixing the exchange rate at a certain level. In the event that the exchange rate is fixed, the external demand shock is entirely passed on to the domestic inflationary environment. Managing this high inflationary environment would require higher interest rates, significantly increasing the social and welfare costs associated with achieving price stability. On the other hand, an inflation targeting strategy allows the exchange rate to entirely absorb the external demand shock.

Scenario 1. Economic Bubble

Figure B.1.1 Real GDP Index and Trend

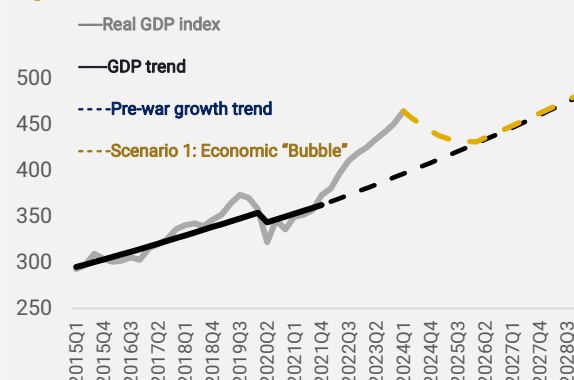
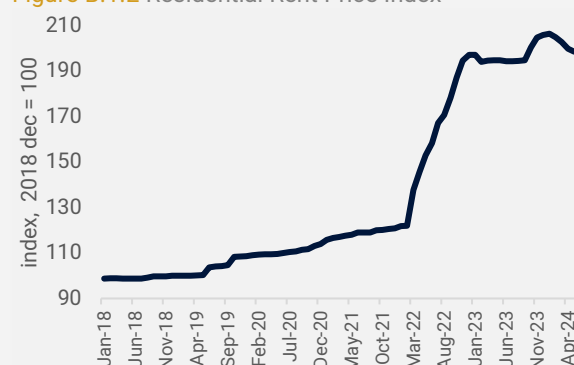


Figure B.1.2 Residential Rent Price Index

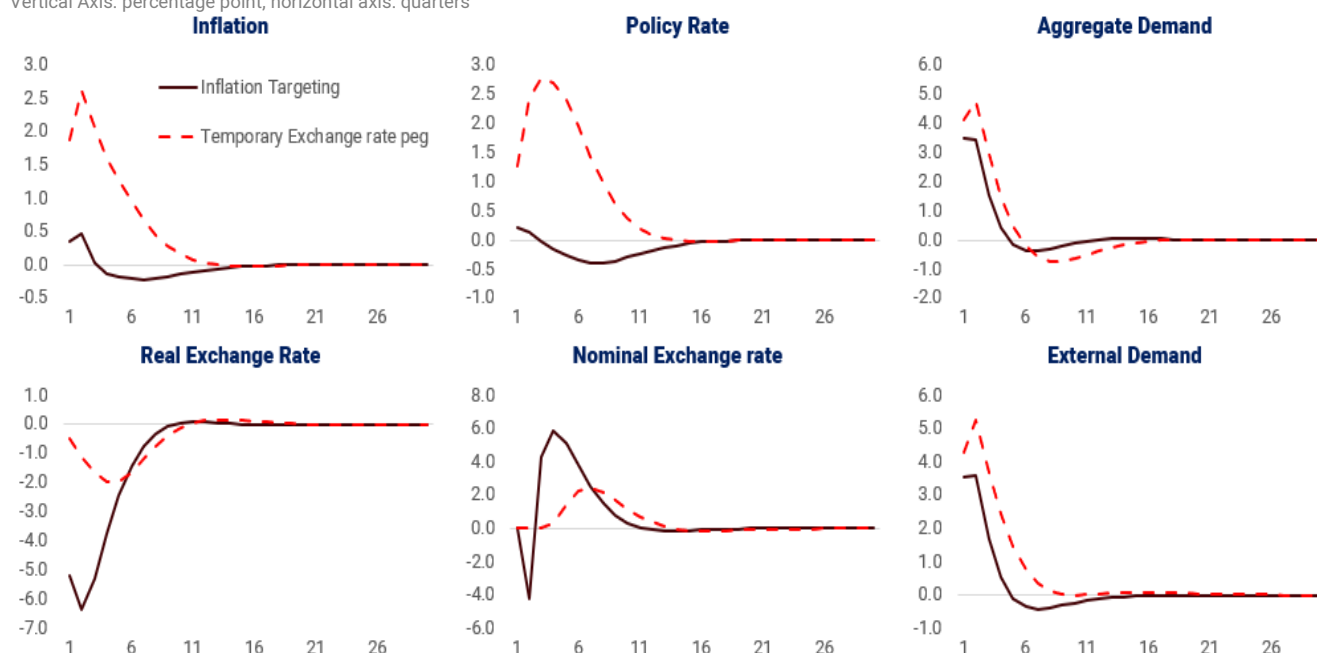


Source: SCRA; CBA calculations

Response of Economic Variables to the External Demand Shock, Assuming Either an Inflation-Targeting Framework or a Temporary Fixing of the Exchange Rate

Figure B.1.3 Deviation of Economic Variables from Equilibrium Levels

Vertical Axis: percentage point; horizontal axis: quarters



Scenario 2. “Low-Value” Productivity Growth

Labor, capital, and financial inflows to Armenia since 2022 have created favorable opportunities in various sectors of the economy. From the perspective of the demand and inflationary environment, this would represent a neutral scenario that does not require a quick correction of the economy. However, in this scenario, opportunities to “easily” generate incomes in certain exportable sectors of the economy result in a redistribution of economic resources (labor and capital) away from other productive sectors of the economy and result in changes in the structure of the economy. In this scenario, structural shifts would take place toward “low productivity” sectors (Figure B.2.5), such as, for example, the financial sector (as has been the case with Cyprus), tourism (e.g. Greece, Georgia, etc.), transit services (e.g. Panama, Gibraltar, etc.). Though these shifts would support meaningful income and GDP growth in the short and medium term, over a longer-term horizon, increased competition for resources would limit opportunities for growth in high-productivity sectors that play an important role in driving strong, sustainable economic growth in the economy. In this context, the role of sectoral policies would be critical, where the primary policy dilemma lies in facilitating the development of the “desired” economic structure over the long run, while refusing opportunities for high growth in the shorter term.

In this scenario (Figure B.2.5), monetary policy would have a neutral impact on structural changes in the economy, while fiscal authorities could use sectoral policies to steer resources toward more productive, exportable sectors, supporting greater economic opportunities over the long term. In this type of scenario, the appreciation of the currency would take place in parallel with non-widespread and sectoral growth in productivity. While this would not necessarily be problematic from a macro perspective, in the context of the aforementioned structural characteristics of the economy and competition for resources, this could pose challenges for specific sectors in the economy.

Scenario 2. Low-Value Productivity Growth

Figure B.1.4: Real GDP Index and Trend

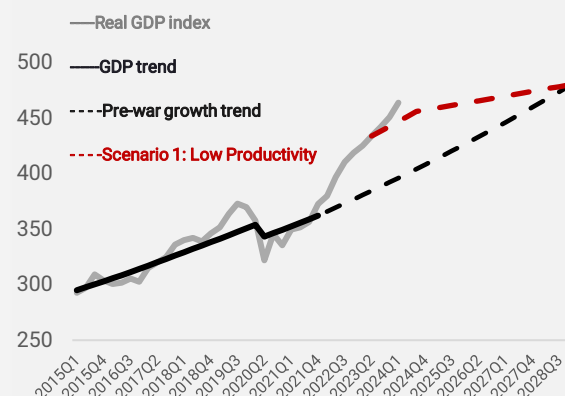


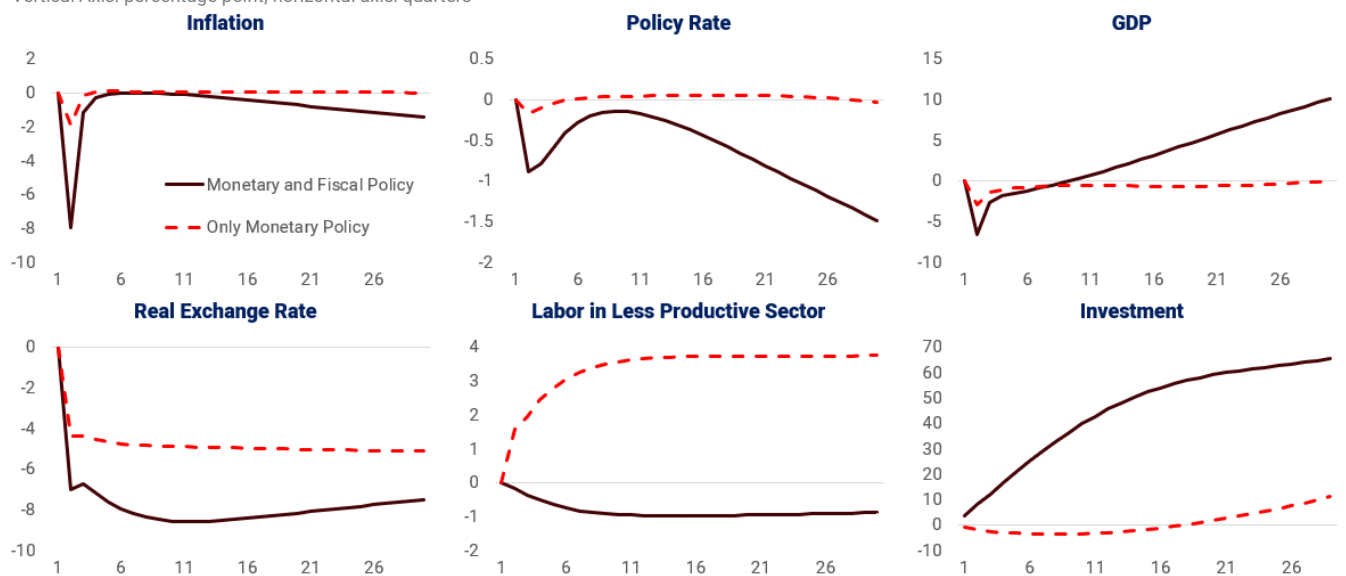
Figure B.1.5: Low-Productivity Employment as a Share of Total Employment



Response of Economic Variables to the External Demand Shock and Economic Restructuring toward Less Productive Sectors. Fiscal Policy Assumes an Optimal Redistribution of Resources toward More Productive Sectors of the Economy

Figure B.1.6: Deviation of Economic Variables from Equilibrium Levels

Vertical Axis: percentage point; horizontal axis: quarters



Scenario 3. High Productivity Growth over the Long Term

Following the onset of the Russia-Ukraine conflict and the emergence of new geo-economic realities, the large labor, capital, and financial inflows to Armenia and strong export demand for Armenian goods and services have created meaningful new opportunities have emerged for Armenia to significantly enhance its economic potential and support high, stable, long-term growth. Specifically, between 2022 and 2024, approximately 13,000 firms and entrepreneurs (capital) have moved to Armenia, of which the majority (55%) are in the IT sector. During this time, the number of registered non-resident employees has increased by approximately 15,000 (Figure B.2.8), the bulk of which took place during 2022. At the same time, the emergence of more favorable competitive conditions for Armenian goods in the Russian market could support expanding production in “traditional” exportable sectors (e.g. alcohol, food production, and so on). Taken together, this could create real opportunities for long-term investments in high-productivity sectors, significantly enhancing the Armenian economy’s long-term potential and growth opportunities. In this context, the primary challenge for economic policies lies in enacting targeted programs aimed at improving the structural and institutional environment, with the ultimate aim of directing “easy” resources toward highly productive and promising sectors and “capitalizing” on these opportunities to facilitate sustainable, long-term growth (e.g. Ireland).

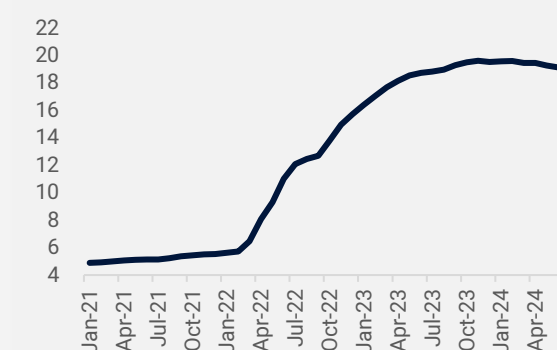
In this scenario of high productivity growth, the real equilibrium exchange rate would appreciate in parallel to growth in export capabilities, contributing to a lower neutral rate in the economy. In contrast to scenario 2, this appreciation would be a natural consequence of widespread growth in productivity and incomes, and would not create any pressures from the perspective of macro stability or the long-term structure of the economy.

Scenario 3. High Productivity Growth

Figure B.1.7: Real GDP Index and Trend



Figure B.1.8: Number of Registered Non-Resident Employees in Armenia
Thousands of persons

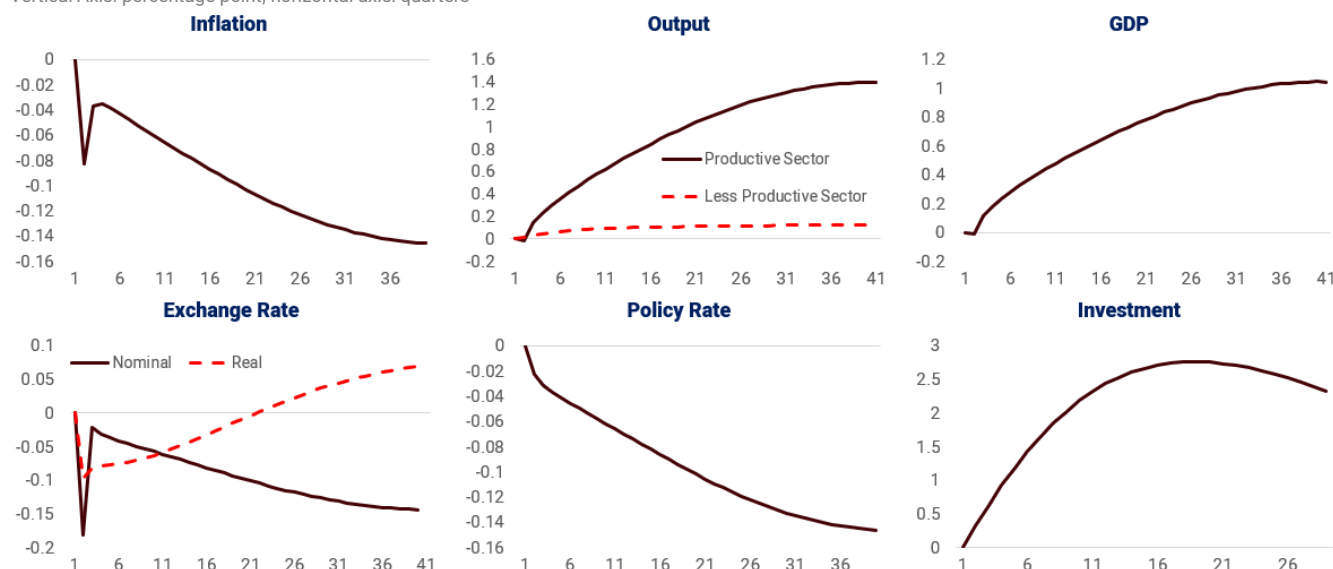


Source: SCRA; CBA calculations

Response of Economic Variables under a High Productivity Growth Scenario

Figure B.1.9: Deviation of Economic Variables from Equilibrium Levels

Vertical Axis: percentage point; horizontal axis: quarters



C. Labor Market and Inflation

Labor Supply and Demand

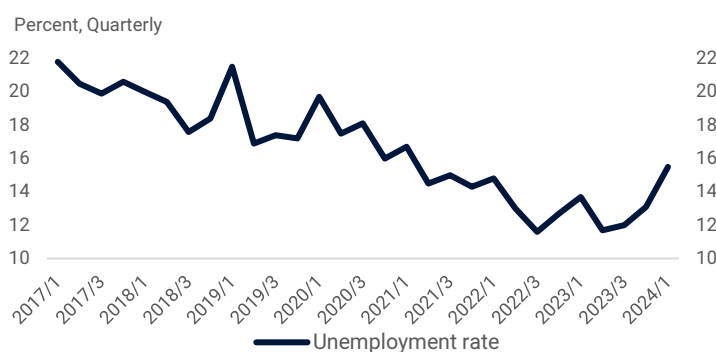
The unemployment rate (derived from SCRA household surveys) increased somewhat from historically low levels in previous quarters, reaching 15.5% in 2024Q1 (figure 3.C.1), reflecting a gradual weakening of labor market conditions. It should be noted that while there was an observed increase in working age population in RA during the same period, the majority of these individuals were classified as part of the economically inactive population. Hence, the uptick in unemployment rate is rather a consequence of the increase in numbers of unemployed rather than the increase in labor supply. The number of registered employees (based on State Revenue Committee data), meanwhile, has continued to increase during the first half of 2024, which could be interpreted as reflecting sufficient labor demand in the market, and could signal uncertainties about labor market conditions, as surveys and data from businesses provided mixed signals about labor market developments. An increase in the number of registered workers may also reflect structural changes within the economy, specifically the reallocation of labor from the informal to the formal sector.

Main uncertainties regarding the labor market outlook surround the potential for an outflow of migrant workers from the Armenian labor market; the scale and pace of the integration of forcibly displaced persons from Nagorno-Karabakh; as well as slowing flows of Armenian labor migrants to Russia and their greater participation in the domestic labor market. In particular, the key risks are centered on the decrease in foreign labor inflows and possible outflows which could reduce the labor supply, hindering long run potential economic growth. A decrease in the number of registered non-resident workers was already observed in the first half of in 2024.

Additionally, the continued narrowing wage gap between Armenia and Russia (especially in the construction sector) could cause flows of Armenian migrant workers to Russia to slow, which would have a positive effect on expanding labor supply domestically as well as improving potential output to some extent (Figure 3.C.3). One interpretation of the meaningful decline in remittances from Russia in recent months (Figure 3.B.5) could be that this already reflects slowing trends in flows of Armenian migrant workers. At the same time, the integration of displaced persons from Nagorno-Karabakh in the economy could have positive impact on labor supply and potential growth. In the short term, this could lead to a modest increase in unemployment. However, in the medium to long term, positive impact on potential GDP growth through increased labor supply and capital accumulation would be expected. Such a scenario would result in a moderation of inflationary pressures stemming from the labor market. Given that the substantial increase in labor resources over first quarter was not accompanied by significant rise in labor supply, the key uncertainty relates to the pace of their integration into the labor market.

Gradual Easing in the Labor Market Supports the Softening in Wage Pressures, but Without a Significant Increase in Labor Supply

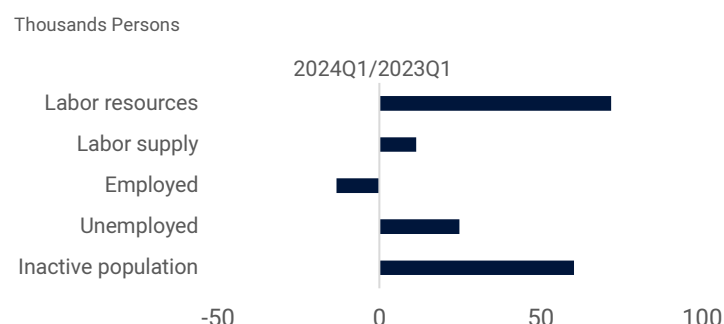
Figure 3.C.1: Unemployment Rate, Armenia



Source: Statistical Committee of the Republic of Armenia

The substantial increase in labor resources raises uncertainties about the pace of their integration into the labor market.

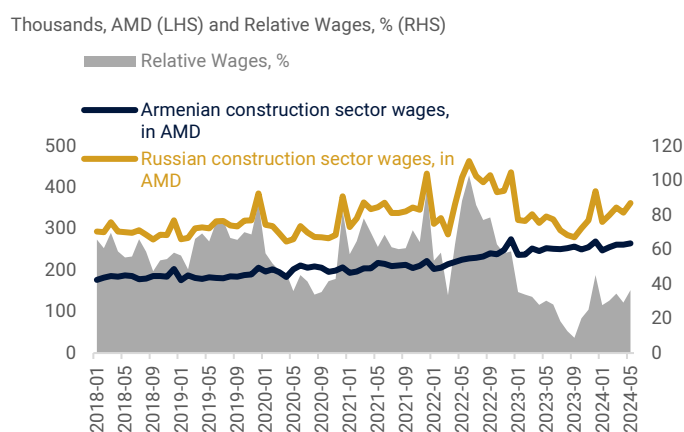
Figure 3.C.2: The Change in RA Labor Resources, 2024Q1/2023Q1



Source: Statistical Committee of the Republic of Armenia; CBA

Sustained Narrowing of Gap between Construction Wages in Armenia and Russia Could Impact Flows of Seasonal Migrant Workers from Armenia to Russia

Figure 3.C.3: Construction Wages in Russia and Armenia



Source: Rosstat; Statistical Committee of the Republic of Armenia; CB

Wage Growth

Private nominal wage growth continued to decelerate in 2024 Q2, reaching 4% y/y and 4.1% in July 2024. (Figure 3.C.4). Nevertheless, the growth in productivity, labor supply, and employment in recent quarters has continued to contribute to weakening inflationary pressures from the labor market, though signals from the labor market are somewhat inconsistent. While wage growth in financial and insurance services was very robust (on the back of strong economic activity) and contributed meaningfully to overall private wage growth, wage growth in many other sectors has continued to adjust downward. The uneven distribution of wage growth across various sectors and the continuous adjustments in others suggest a notable reduction or neutralization of labor market pressures. Additionally, a potential increase in labor supply could further ease labor market conditions and reduce market-driven inflationary pressures in the medium to long term.

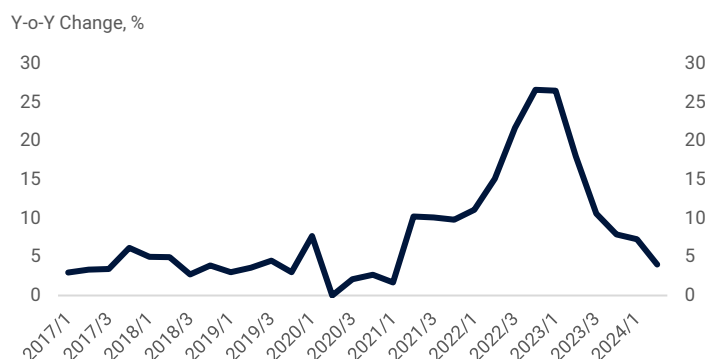
Inflation

In Q2 2024, following periods of deflation episodes, the 12-month inflation has somewhat increased (1.3% Y-o-Y in August 2024), primarily supported by seasonal food price increases. The subdued inflationary pressures are primarily driven by weakened pressures coming from the external sector, the impact of the monetary policy implemented in recent years, the gradual easing of labor market conditions and some reduction in aggregate supply and demand imbalances. External and domestic demand-driven services inflation also continued to soften. Meanwhile, still higher domestic demand, coupled with revisions to the tax code (namely excise tax, taxi services tax, etc.), as well as the planned review of public transportation fares in Yerevan starting January 2025 may continue to pose some inflationary risks. The latter, however, is estimated to have limited overall effect on inflation (max. one-time effect of around 0.5 p.p.), given its relatively small share in the consumer basket. Moreover, upward risks to oil prices, as well as the observed slight monthly increases in imported food prices in recent months, could cause imported deflationary trends to weaken considerably or reverse.

Core inflation, which excludes seasonal food and regulated services, has also shown a recent uptick, standing at 0.3%, Y-o-Y in July 2024 and largely reflecting sustained declines in imported food and non-food product prices (e.g., bread, oils & fats, sugar, etc.), although imported food prices are exhibiting modest inflationary pressures on a monthly basis. Meanwhile, Non-Traded Sticky Price Inflation (NTSPI) somewhat moderated in July 2024, reaching 2.4% Y-o-Y and revealing moderation and adjustment of aggregate demand.

Increases in Labor Supply Driven by Labor Resource Expansion Could Contribute to Weakening Inflationary Pressures, though Signals from the Labor Market are Somewhat Incongruous

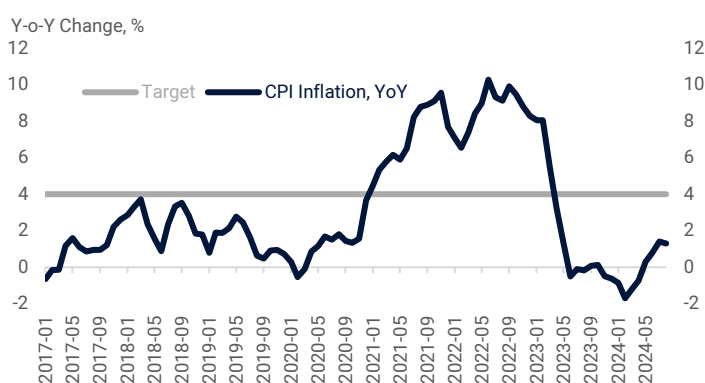
Figure 3.C.4: Nominal Private Wage Growth, Armenia



Source: Statistical Committee of the Republic of Armenia

A Subdued Inflationary Environment is Observed, Driven by the Diminishing Influence of Base Effects and Continuous Deflationary Pressures from the External Sector.

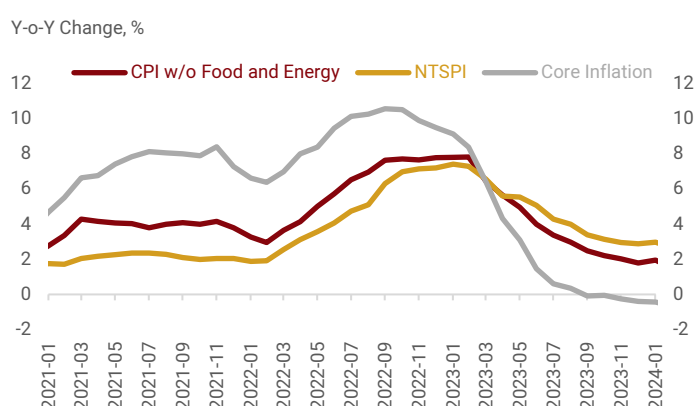
Figure 3.C.5: Y-o-Y CPI Inflation and the Target, Armenia



Source: Statistical Committee of the Republic of Armenia, CBA

Non-Traded Sticky Prices Have Largely Stabilized

Figure 3.C.6: Underlying Inflation Measures, Armenia



Source: Statistical Committee of the Republic of Armenia, CBA

Inflation (cont.)

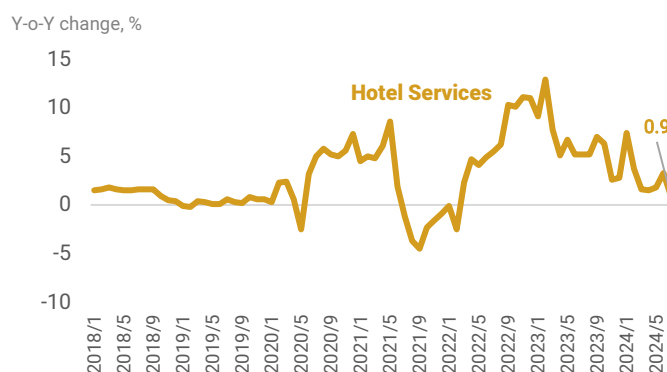
The uptick in momentum observed in Q4 2023, driven by high demand for residential rent and other non-tradeable sticky services, has gradually faded away in the first half of 2024, suggesting that upward pressures on sticky prices have largely dissipated. At the same time, inflation for services that are highly exposed to external demand, such as air transport, restaurants, hotels, and recreation and culture, has meaningfully softened. Against this background, third-party projections of inflation through the rest of 2024, including the IMF and surveys of market participants, suggest headline inflation in the near-term is likely to remain at or below target (Table 3.C.1).

Inflation Expectations

According to the results of the household inflation expectations survey conducted by the Central Bank of Armenia, in the third quarter of 2024, a higher number of households expect a relatively rapid increase in inflation compared to the previous quarter. This is somewhat consistent with the inflation within a positive range in recent months, although it remains well below the target level. At the same time, uncertainties regarding inflation continue to persist at a high level and are increasing, which likely reflects public perceptions about the persistence of high inflation for certain goods. It is important to note that the overall trends in the third-quarter survey, despite some decrease in inflation expectations in the previous quarter, suggest that the adjustment of inflation expectations could occur much more rapidly, potentially creating risks in the context of overall inflation acceleration.

External Demand-Driven Inflationary Pressures Have Eased

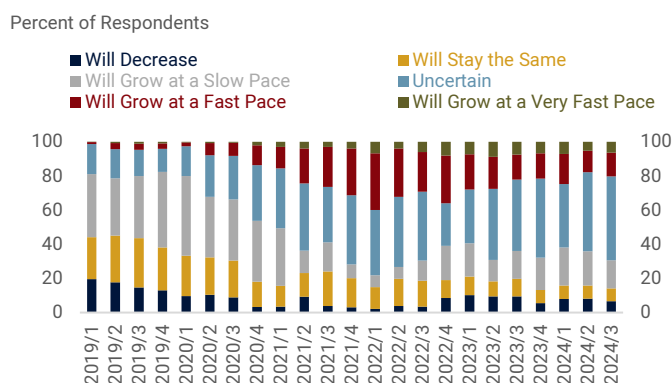
Figure 3.C.7: Hotel Sectors Inflation



Source: Statistical Committee of the Republic of Armenia, CBA

Larger Proportions of Respondents Expect Higher Inflation than in the Past and Uncertainty Remains Elevated

Figure 3.C.8: Survey on Households' Inflation Expectations, Armenia



Source: CBA

D. Financial Markets

Market Expectations of Policy Rates

According to the latest survey conducted by the CBA among financial market participants (including commercial banks, institutional investors, asset managers, and others) as of September 6, 2024, market participants on average expect the CBA to continue gradually decreasing the policy rate over the course of the next eight decisions in line with the expected macroeconomic outlook. The key risks and issues that respondents emphasized as possible uncertainties related to these projections have served as an important input into the taxonomy of scenarios presented in Section E.

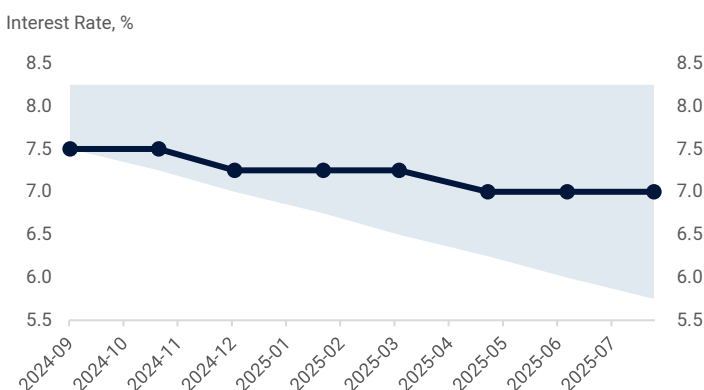
According to market expectations derived from the yield curve, bond markets reflect slightly higher interest rate expectations compared to surveys over the one-year horizon, though the difference between expectations of surveys of market participants have declined compared to previous quarters. Overall, since the June policy rate decision, the longer-term edge of the yield curve hasn't changed dramatically indicating potentially unchanged inflation expectations and perceived country risk premium and convergence of the policy rate towards the long run neutral level (see Figure 3.D.3) over the long term horizon, while the shorter-term portion of the curve has adjusted downwards following MP decisions.

Neutral Interest Rate

Although uncertainties persist, however the recent actions of CBA in reducing the refinancing rate significantly reduced the probability of policy being too tight, given the weak inflationary environment. Recent uptick in country risk premium (Figure 3.D.6) and geopolitical uncertainties in the region could indicate that neutral rate in Armenia is somewhat higher. Additionally, elevated long term interest rates globally¹ might suggest that underlying equilibrium interest rates in the world are higher, thus necessitating an upward revision in neutral rates in emerging market economies. At the same time, growth of potential and export capacities in Armenia in recent years and appreciation of real effective exchange rate might suggest a lower underlying interest rate in domestic economy, as investors also consider long-term domestic currency dynamics when making cross-border investment decisions.

Most Market Participants Expect Gradual Declines in Policy Rate over the Next Year

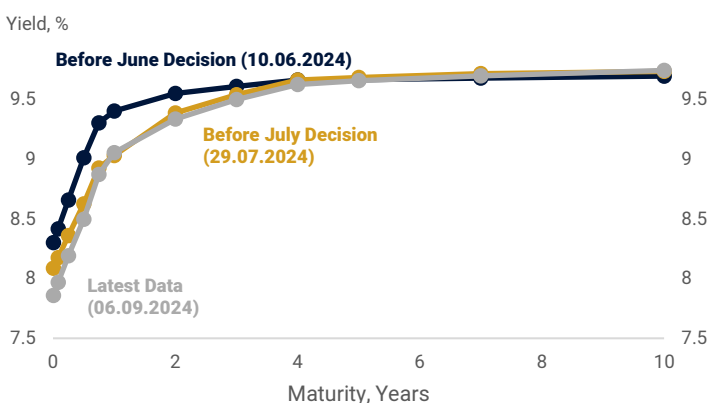
Figure 3.D.1: Financial Market Participants' Expectations of the CBA Policy Rate According to September 6, 2024 CBA Survey



Source: CBA

Both Short- and Long-Term Bond Yields Continue to Decline

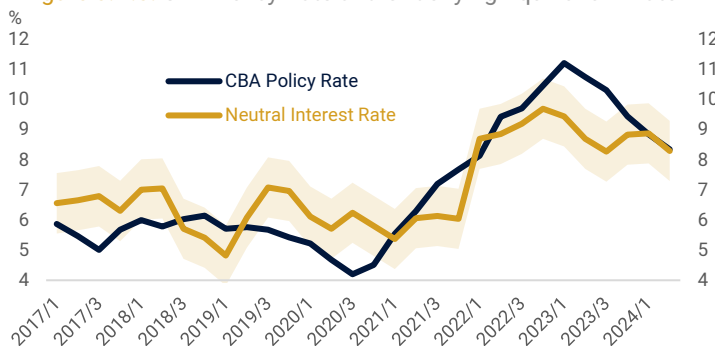
Figure 3.D.2: Yield Curve Dynamics, Armenia



Source: CBA

How Tight are Monetary Conditions?

Figure 3.D.3: CBA Policy Rate and Underlying Equilibrium Rate



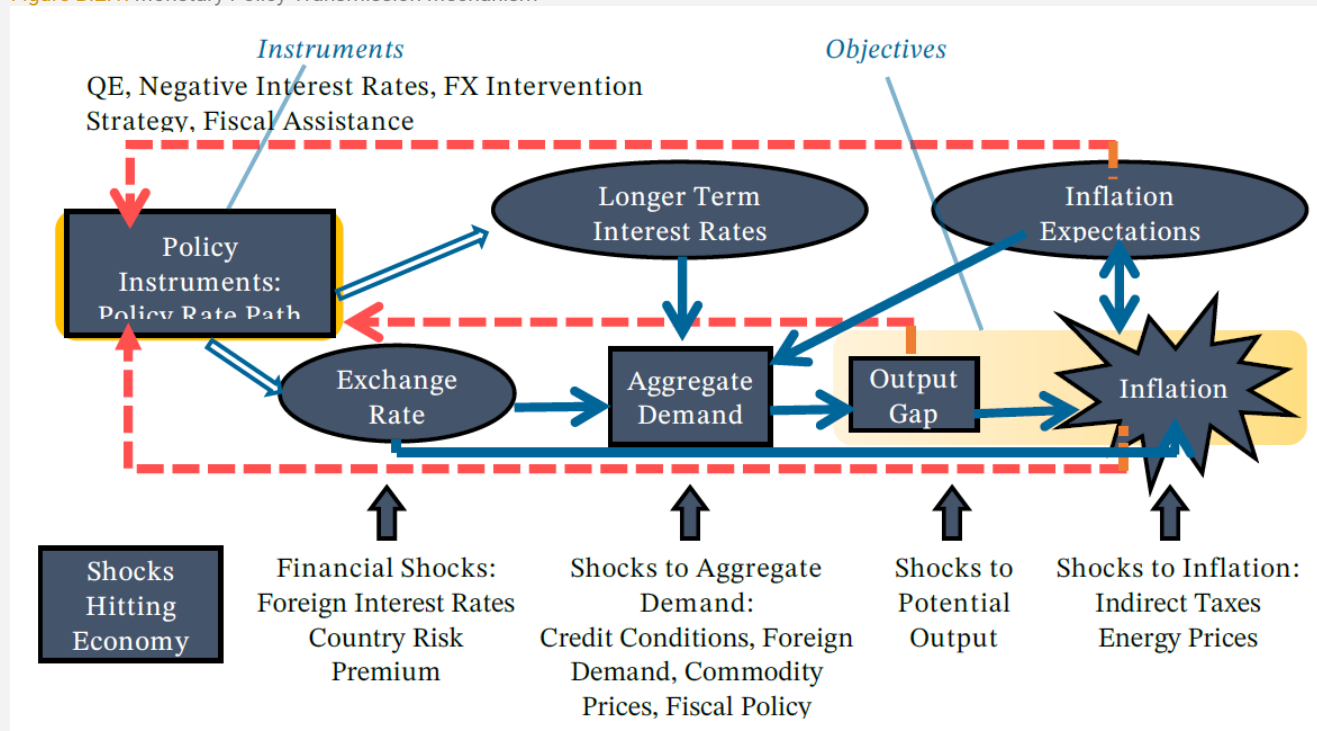
Source: CBA

¹ There is also considerable diversity in views and uncertainty surrounding neutral rates globally and particularly in the United States (see Box 2 in 2024Q2 MPR).

Box 2. Transmission Mechanism of Monetary Policy

Monetary policy involves the process of setting policy instruments, while the monetary transmission mechanism focuses on how these instruments impact the economy. Understanding the transmission mechanism is crucial in assessing how policy affects the overall economy. Central banks operating under a flexible inflation-targeting regime are committed to adjusting the policy rate aggressively enough to anchor inflation and inflation expectations. Usually, the primary channels through which monetary policy operates are the interest rate, exchange rate, and credit channels, as illustrated in Figure B.3.1.

Figure B.2.1: Monetary Policy Transmission Mechanism



Source: Laxton, Galstyan and Avagyan, Book on "Prudent Risk Management Approach to Monetary Policy", Unofficial working Draft

The interest rate channel is a crucial component of monetary policy transmission. The expected path of the policy rate significantly impacts long-term interest rates. When the central bank sets short-term rates, it influences interbank rates, which in turn affect borrowing and lending rates. Communication regarding the expected policy rate path is vital as it directly influences long-term rates, thereby influencing the decisions of households and firms. Additionally, central banks may use unconventional tools, like quantitative easing, to influence long-term rates when short-term rates are constrained by the effective lower bound.

The exchange rate channel is particularly important for small open economies like Armenia. Changes in interest rates affect the returns on financial assets, influencing the exchange rate. A higher policy rate typically makes domestic currency bonds more attractive, leading to currency appreciation. This, in turn, affects trade balances and inflation by altering import and export prices. This channel becomes even more crucial in economies where a significant portion of domestically consumed goods are imported. In such contexts, currency fluctuations have a direct and pronounced effect on inflation, as changes in the exchange rate swiftly translate into changes in the domestic prices of imported goods. This underscores the heightened sensitivity of these economies to external shocks.

Inflation expectations, which depend on the credibility of the central bank, are another critical aspect of transmission mechanism. If inflation expectations rise in response to economic disturbances, they can directly drive inflation as businesses adjust prices and consumers alter spending in anticipation of higher prices in the future. Anchoring these expectations simplifies the task of achieving price stability.

Lags in transmission mechanism: The effects of monetary instruments on the economy can vary. While some believe that the transmission mechanism operates with a lag, this is not always the case. If the central bank operates with a well-established and transparent framework, financial markets may adjust their expectations accordingly, even before any policy actions are taken. For instance, in 2022, when the Federal Reserve indicated that inflation was not transitory, financial market interest rates rose significantly in response to this communication, as shown in Figure B.3.2. This example clearly illustrates how the communication can lead to immediate adjustments in financial markets, even before formal policy changes are made. Lags in the transmission mechanism typically refer to the time delay between changes in market interest rates and their impact on the real economy and inflation.

When financial markets are well-acquainted with a **central bank's framework, principles, long-term objectives**, and policy priorities, they can rapidly adjust their expectations in response to key economic data releases or major news. This adjustment can occur even before any formal policy rate changes are made, or before the central bank has communicated its intended response to the changes. Such responsiveness underscores the strength of a well-established and transparent monetary policy framework, where market participants can accurately anticipate the central bank's actions, thereby enhancing the immediacy and effectiveness of the policy transmission mechanism. A recent example of this dynamic can be observed in the financial markets' swift response to significant data releases related to labor market in the United States. Understanding the Fed's dual mandate has led markets to anticipate a quicker reduction in the policy rate in response to a weakening labor market (Figure 3.A.3).

Continued on the next page

Link between short and long term interest rates: Understanding the term structure of interest rates and the link between short and long term interest rates is vital in assessing the effectiveness of monetary policy transmission mechanism. It is a common misconception that if policy rate adjustments do not significantly influence long-term rates, such as those for loans or mortgages the effectiveness of monetary policy transmission is thereby questioned. However, this interpretation can be misleading. The policy rate primarily affects the cost of short-term financing and is thus more directly correlated with short-term interest rates, such as short-term bond rates or deposits. The cost of short-term liquidity for banks has a limited direct effect on long-term interest rates. What matters more is the expected path of the policy rate. Let's imagine a credible central bank responding aggressively to economic shocks, establishing short-term adjustments that reinforces long-term stability—such as keeping inflation expectations well-anchored—the short-term rates may rise substantially, while long-term rates may exhibit only minor changes. This case demonstrates effective monetary policy transmission, contrary to the notion that a lack of significant movement in long-term rates indicates a weak pass-through.

For instance, during the 1980s, the Federal Reserve, under Paul Volcker, sharply raised interest rates to combat double-digit inflation. At that time, the lack of credibility and limited monetary policy transparency kept long-term interest rates elevated due to high inflation expectations. Understanding the expectations theory of the term structure was crucial; had markets believed that future inflation would decline, long-term rates should have been lower. This historical context underscores the importance of the central bank's role in managing expectations to ensure the effectiveness of the transmission mechanism.

While long-term interest rates should generally adjust in response to changes in short-term borrowing costs, **factors that determine long-term interest rates** are crucial for assessing the monetary policy transmission. Competition among commercial banks, inflation expectations, demand for financial products, perceived consumer risk, economic outlook, and country risk premium all play key roles in shaping the pricing of financial products. These drivers can sometimes obscure the direct link between policy rates and long-term rates. In Armenia, during monetary tightening or loosening cycles, long-term rates such as for deposits and loans have generally aligned with policy rates, as shown in Figure B.3.4. However, for example during the post-COVID period, this relationship appeared somewhat distorted. As the CBA began lowering rates in early 2020 in response to the economic impact of COVID-19, long-term rates declined only slightly, reflecting heightened uncertainty and risk perceptions about the future.

Another factor distorting the link between policy rates and longer-term rates, especially over the current monetary loosening cycle, is the government's mortgage interest rate subsidies program. This policy reimburses the entire amount of mortgage interest payments from income taxes if the mortgage is used to purchase a newly built property. Essentially, the government pays back the borrower's income tax in an amount equivalent to the mortgage interest payments, making borrowers almost insensitive to mortgage interest rates. The policy's impact has been substantial, with mortgage lending volume increasing significantly since its inception. This elevated demand for long-term funds further contributing to this "temporary disconnect" and adding a layer of complexity to the monetary policy transmission in Armenia.

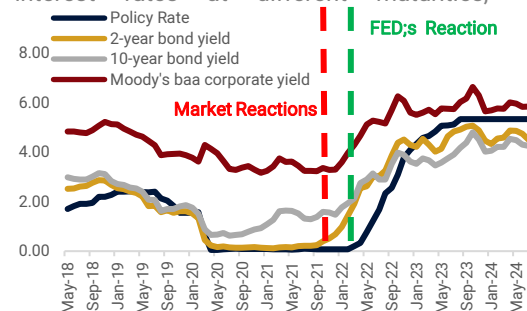
Challenges in emerging market economies: Emerging economies often face unique challenges in effectively transmitting monetary policy. High financial market volatility, reliance on foreign capital, and underdeveloped financial systems can undermine the stability and impact of monetary actions. Structural issues, such as weaker institutions and less credible policy frameworks, further complicate policy transmission. In Armenia, despite significant progress by the Central Bank in refining its monetary policy framework, challenges persist due to the country's position as a small open economy in a volatile region. The global shocks following the COVID-19 pandemic and the Russian-Ukrainian war have placed additional pressure on central banks.

The steady decline in **dollarization**, from about 60 percent to 30-40 percent over the last years, reflects growing confidence in the local currency and the Central Bank's credibility. This decrease should have positively affected the monetary policy transmission mechanism, especially the exchange rate channel, contributing to greater economic stability. However, dollarization remains a challenge, complicating the link between domestic monetary policy and broader economic outcomes.

In conclusion, despite numerous constraints and uncertainties, the CBA has successfully achieved its price stability goal. Since adopting an inflation-targeting framework in 2006, average inflation has been too close to 4.0%, matching the target. The Central Bank's flexible approach, which considers various perspectives, has been crucial in managing inflation and anchoring expectations. Additionally, strengthening the credit channel by fostering confidence in the financial system has been essential in enhancing the overall effectiveness of monetary policy.

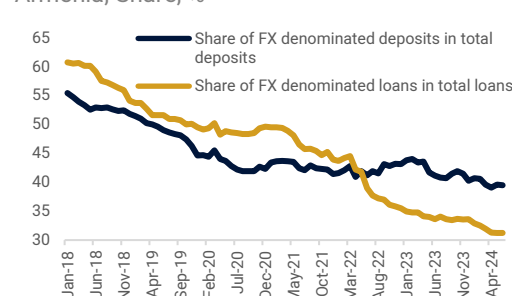
Market Interest Rates Rising Before the Effective Fed Funds Rate Lift-Off in response to FED's communication and economic data release

Figure B.2.2: Effective FED funds rate and market interest rates at different maturities, %



High level of dollarization in emerging market economies can weaken the exchange rate channel

Figure B.2.3: Loan and deposit dollarization in Armenia, Share, %



Longer term interest rates, such as loan, deposit rates have more complex structure than simply reacting to short term interest rates

Figure B.2.4: Policy rate and market interest rates in Armenia, basis points

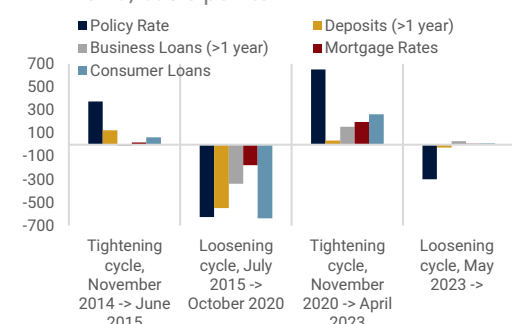
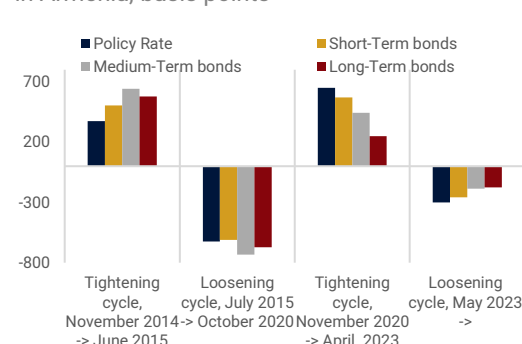


Figure B.5: Policy rate and market interest rates in Armenia, basis points



Source: CBA

Bank Lending Conditions

CBA surveys show that the index of commercial bank lending conditions continued to follow a net easing trend across all types of loans, while interest rates remain relatively stable, not yet showing signs of easing due also to a net increase in demand for these loans.

The funds attracted and provided by commercial banks continued to grow in Q2 2024, likely reflecting high economic activity in the real sector. Loans provided to households continue to grow strongly reflecting high, yet slowing, demand for housing. Despite persisting uncertainties growth in business loans rebounded in Q2. Still high GDP growth, accumulated savings in the private sector, as well as a relatively lower debt burden (relative to incomes) in the private sector could serve to drive additional consumer confidence, which could motivate further growth in credit fueling domestic demand driven pressures. On the other hand geopolitical uncertainties in the region might deteriorate household and business confidence thus curbing economic activity.

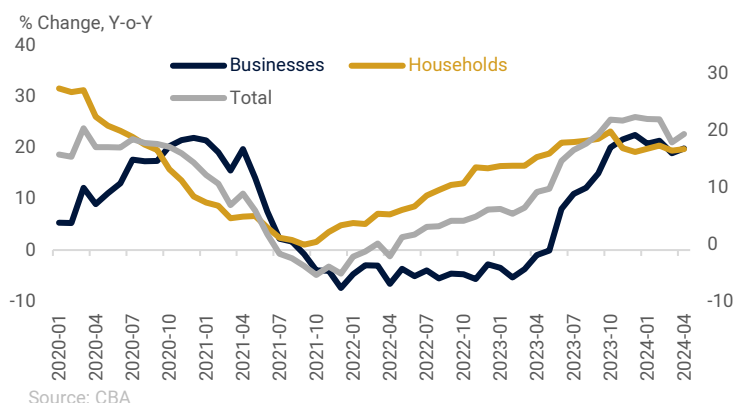
Nominal Exchange Rate

Since the beginning of 2024, the dram has held relatively steady versus the US dollar, Euro, and Ruble (Figure 3.D.5). However, very modest appreciation of the dram has been observed relative to Ruble in recent weeks, as the ruble weakened due to strong domestic demand pressures and elevated inflation expectations in Russia (Section 3A). The steady dram is a reflection of both appreciation and depreciation forces in the domestic economy as well as the rest of the world. The key risks and uncertainties include the possibility of sustained growth in exports and trade, weaker imports of consumption goods, as well as stronger external demand for domestic services (which, though softened, are still at high levels), to generate appreciation pressures. On the other hand, these trends could be offset by uncertainties in the region and further increase in country risk premium as well as the possibility of domestic demand proving to be stronger, with greater domestic consumption (driven by accumulated savings and room for growth in consumer lending) driving higher imports.

The country risk premium (Figure 3.D.6; estimated as the difference between the yield on Armenian Eurobonds maturing in 2029 and a benchmark risk-free interest rate with the same maturity) somewhat increased in Q3 2024, after a steady decline during the previous quarters. The slight increase in Armenia's country risk premium reflects the geopolitical uncertainties in the region, rather than country specific factors related to Armenia. Overall emerging market economies continue to experience a lower risk premium due to less concerns for capital flows from emerging market economies to advanced economies as the recent data in US indicated a weakening of labor market and significantly reduced the probability of "higher for longer" policies in US.

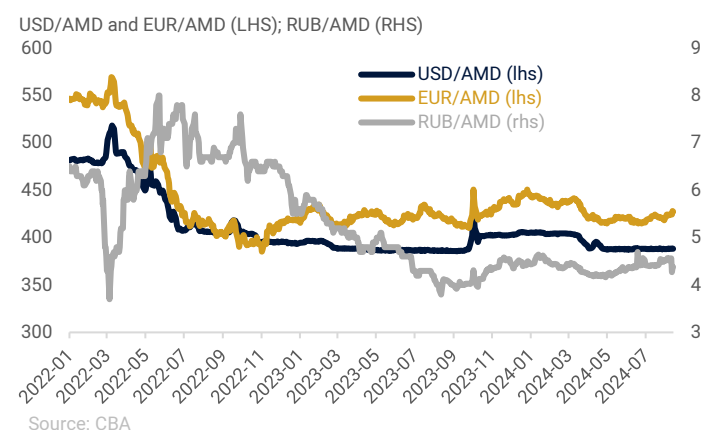
Demand for Consumer Loans Continue to Grow

Figure 3.D.4: Change in Total Loans Provided by Commercial Banks to Households and Businesses, Armenia



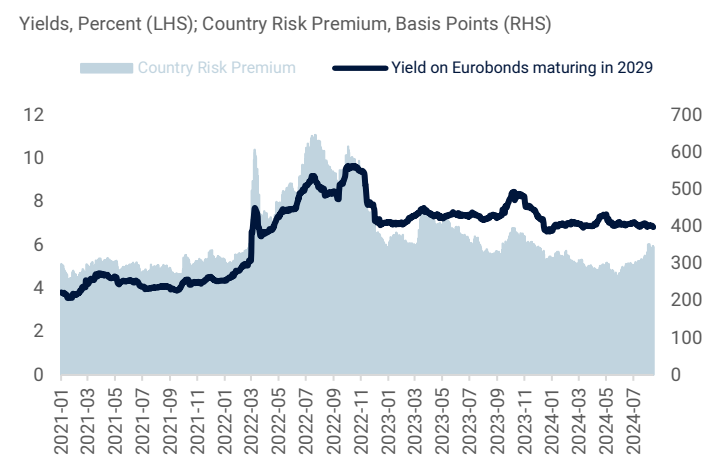
Significant Appreciation of AMD since mid-2022

Figure 3.D.5: Nominal Exchange Rate for AMD against USD, EUR and RUB (Lower Value means Stronger Dram)



While the Country Risk Premium Continues to Decline, Uncertainty Regarding the Outlook for Regional Geopolitical Tensions Still Poses Upside Risks, with Implications on Financial Markets

Figure 3.D.6: Yield on Armenian Eurobonds Maturing in 2029 and Estimated Country Risk Premium



Box 3. Scenarios-Based Forward Guidance: The Role of Market Expectations

Approaching monetary policy through the lens of risk management under FPAS Mark II requires a different approach to the scenario-building process. Rather than treating its own central projections as the best piece of information and forcing market participants to coalesce around this singular view of unfolding economic conditions, the CBA instead utilizes market expectations for the central bank policy rate as a starting point for policy deliberations. Market expectations, by design, contain significant information about market views on current economic developments and possible behavior of monetary policy in the future economic environment.

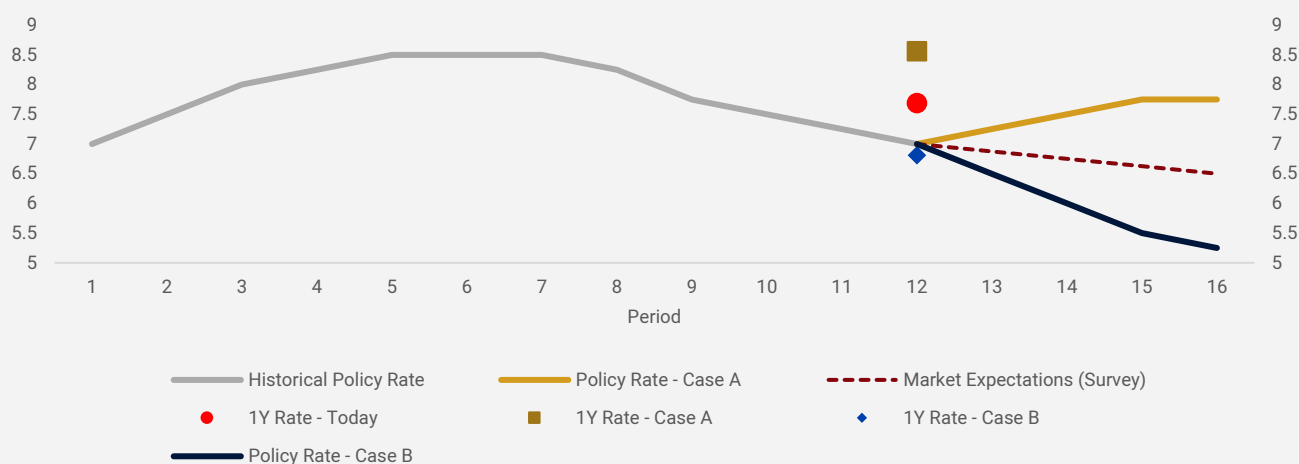
Before the development of flexible-inflation-targeting approaches, monetary policy communication predominantly focused on direct actions like the setting of short-term policy rates. However, short-term rates are not particularly relevant for most people; longer-term interest rates tend to hold greater significance as these are the types of maturities at which households and businesses typically borrow and lend. The two principal components that make up longer-term rates are the term and risk premiums and the expected path of the policy rate. Of these two components, the expected path of the policy rate is most significant. As inflation targeting regimes were adopted, transparency in communicating a central bank's perspective and rationale became a necessary precondition for the regime's efficacy. With greater central bank clarity, financial markets gain the agility to recalibrate efficiently between policy decisions. Importantly, on decision dates, policymakers do not merely set the current policy rate; they also provide insights into their monetary policy decisions, including how the policy rate may change in the future based on different narratives, scenarios, and assumptions.

In advanced economies with well-developed financial markets (including secondary bond markets), gleaned market expectations of the future path of the policy rate is a relatively straightforward task. For countries with less developed financial markets that lack options markets, such as Armenia, the task is less straightforward. It requires synthesizing multiple approaches, without resorting to "over-modeling" and attempting to ascribe a false degree of precision that is based on unreliable or incomplete information.

Market Expectations of the Policy Rate: We synthesize two approaches. First, we take as a starting point information from the spot market for shorter-term bonds (one year and less) and perfect foresight-based inferences about the term structure of the yield curve. We recognize that data from the spot market is imperfect and cannot paint a complete (or always accurate) picture about market expectations of the policy rate. For example, bond markets experience high term premium volatility, face frequent liquidity and other shocks, and so on, making it difficult to perfectly assess which price movements result from changes in expectations versus other factors. Therefore, we dovetail this approach with a high-frequency survey of policy rate expectations among financial markets (including commercial banks, institutional investors, asset managers, and others). Of course, when considered in isolation, survey results are also imperfect, in the sense that they do not always align with what is actually priced in markets. Therefore, as a rule, we evaluate both survey- and bond market-based measures in order to develop a rich and sensible understanding of market expectations for the future path of the policy rate, without underselling the fundamental uncertainty and room for a range of interpretations.

Scenarios-Based Forward Guidance: To graphically illustrate our case scenarios relative to market expectations, we rely on a simplified approach using the one-year bond rate (see Figure B.4.1). The black line represents the CBA's refinancing rate historically. The one-year bond rate at the time of the decision is depicted with a red dot, representing the market's assessment of the most probable trajectory of the average policy rate over the course of the coming 12 months, plus some compensation for uncertainty regarding inflation and policy rates over this horizon (term premium). This provides a reference point around which the illustrative Case A (orange line) and Case B (blue line) scenarios for the policy rate are constructed. Additionally, we show the corresponding one-year bond rates that would be consistent with the Case A scenario (light orange dot) and Case B scenario (light blue dot) materializing. Finally, the average of the financial market survey responses is shown with the dotted gray line.

Figure B.3.1: Hypothetical Example of Case Scenarios and Market Expectations under FPAS Mark II
Interest Rate, %



Source: CBA

E. Taxonomy of Scenarios

Purpose and Structure of the Taxonomy

The purpose of this taxonomy is to provide a structured assessment of the different types of risks and shocks that can affect the Armenian economy and describe their impact on inflation and economic environment in Armenia, as well as loosely illustrate implied monetary policy reactions by the CBA. Having a structured framework for classifying these risks allows the reader to more effectively understand the wide range of information, ideas, and opinions that are held by various stakeholders, from the Board and Staff of the CBA to financial markets and the public. To allow the diversity of viewpoints among these various groups to flourish and to facilitate the development and communication of better policy decisions, it is imperative to structure these views within a cohesive structure. The taxonomy is intended to provide a concise summary of the Staff's views about the totality of all the policy-relevant risks held by various stakeholders, which are most policy-relevant for Armenia.

Table E.1: Taxonomy Structure

Case A-Type Considerations	Case B-Type Considerations
Expansionary Demand: Shocks result in aggregate demand rising more than aggregate supply, increasing underlying inflation. This requires an increase in the policy rate path that is sufficiently aggressive to anchor inflation and inflation expectations. Complications: credibility is fragile Output + Inflation +	Contractionary Demand: Shocks result in aggregate demand rising less than aggregate supply, decreasing underlying inflation. This requires a decrease in the policy rate path that is sufficiently aggressive to anchor inflation and inflation expectations. Complications: ELB, stuck in low inflation trap Output - Inflation -
Contractionary Supply: Nasty stagflationary shocks that require an aggressive upward adjustment in the expected path of the policy rate, resulting in a significant short-run tradeoff between inflation and other real objectives (e.g., unemployment) Complications: Hard to hit the wounded economy Output - Inflation +	Expansionary Supply: Favorable supply shock(s) reduce inflation toward the target and results in higher growth as well as lower interest rates; is a policymaker's dream. Complications: can be risky if this environment is extrapolated into the future Output + Inflation -

The taxonomy is classified into those that can have Case A-type (expansionary demand, contractionary supply) and Case B-type (contractionary demand, expansionary supply) considerations. Although the classification of shocks can be done in different ways across different features, the one presented here is based on a typical distinction between supply and demand side forces and is particularly useful from policymakers' point of view as it provides a framework for analyzing the source of inflationary pressures and potential risks to the economy. By understanding the nature and magnitude of different types of shocks, policymakers can then design appropriate monetary policy responses to mitigate the negative impact of adverse shocks and leverage the benefits of positive shocks.

Taxonomy of Scenarios: Q3 2024

Table E.2: Taxonomy of Scenarios: Q3 2024

Case A-Type Considerations	Case B-Type Considerations
Expansionary Demand - Recent developments in the U.S. labor market, marked by an increase in unemployment and a reduction in new jobs, have led market participants to expect a sharp decrease in the Federal Reserve's interest rate. However, economic activity in the U.S. remains robust and significant progress has been made in bringing inflation closer to the target level. Despite this, uncertainty persists regarding the long-term neutral interest rate and the eventual return of inflation to the target level. The likelihood that U.S. long-term interest rates will return to pre-COVID levels after the normalization of monetary conditions is quite low, which, in terms of the neutral interest rate for RA, mainly poses risks of maintaining it at a high level. - There are inflationary risks in the U.S. economy due to post-election developments. In particular, an increase in fiscal stimulus is likely, as well as a rise in tariffs between the U.S. and major trading partners, which could create inflationary pressures in international commodity markets. - Continued conflict between Russia and Ukraine and recent escalation amid sustained sanctions against Russia poses risks for further new flows of international visitors to Armenia, representing new waves of external demand for Armenian goods and services that could generate excess demand and inflationary pressures in the Armenian economy, albeit weaker, than in the last cycle. At the same time, the impact of the latter on risk balancing is small, given that the Armenian economy is in an adjustment phase, although it may have some effect on that process. - The high economic growth recorded in the domestic economy may reflect higher levels of productivity, an increase in the marginal cost of capital and a rise in the neutral interest rate. In this case, there are risks of continued acceleration of demand and increased inflationary pressures. - In contrast to the significant reductions in country risk premiums observed in other developing nations, Armenia's country risk premium has remained relatively stable at a low level since the beginning of the year, with a slight upward trend in recent weeks. This trend reflects the unique characteristics of Armenia and the region, including uncertainties surrounding border tensions, regional geopolitical developments, and other factors. In this context, a potential upward adjustment of the country risk premium poses risks related to an upward revision of the neutral interest rate, which, all else being equal, could predominantly result in inflationary pressures given a more accommodative monetary policy stance. - High inflationary expectations continue to remain <i>(Continued on the following page)</i>	Contractionary Demand - Concerns about a weak economic outlook translating to risks of a serious contraction persist in the global economy, especially in the European Union and China. - While growth in Russia remains at high levels, new Western sanctions nevertheless pose concerns to the sustainability of future growth. In this context, any potential sharp slowdown in economic activity could have negative implications on Armenia, including further slowdowns in inflows of remittances to Armenia as well as general contractionary impacts on external demand for goods and services in the Armenian economy. - Continued overheating in the real estate market poses risks to the growth outlook in the construction sector as well as real estate prices, with the potential for deceleration in the construction sector to have negative spillovers on other sectors of the economy, leading to deceleration and domestic demand-driven disinflation. - Slowing wage growth and rising unemployment amid non-widespread and non-inclusive economic growth reflected in highly concentrated income in specific sectors in parallel with relatively muted private consumption and low inflationary environment, drive uncertainty about domestic demand conditions in Armenia and pose risks that a negative demand environment could emerge, with implications for sustained deflationary pressures in the medium term. - Headline inflation remains persistently below the target for a long period, which could lead to a sustained decline in long-term inflation expectations and contribute to restrained consumption patterns. - Risks of additional demand restraint by fiscal policy. - Enhanced tax revenue collection in the face of relative increase of capital expenditures share in the total government expenditure could result in fiscal policy having much more restrictive impacts on aggregate demand. - Additional restraint on demand, along with possible tax administration tightening, legislative changes and other measures needed to improve the tax-to-GDP ratio and finance the budget deficit in the future. <i>(Continued on the following page)</i>

Case A-Type Considerations (continued)	Case B-Type Considerations (continued)
Expansionary Demand (continued) • Sustained excess demand scenarios: ◦ Above-potential economic growth suggests that generalized excess demand conditions persist and may be growing. ◦ Private consumption (and resulting inflationary pressures) could accelerate on the back of accumulated private savings, the continued high growth of consumer loans, strong wage growth and income growth driven by positive developments in the tradable sector. • The highly stimulative stance of fiscal policy is due to the risks of tax revenues falling short of projections and the need to increase current expenditures to implement social assistance programs. On the one hand, this will contribute to the formation of a more stimulating fiscal position; on the other hand, it carries risks of a potential increase in public debt.	Contractionary Demand (continued) • Uncertainty about a potentially lower domestic neutral rate, resulting in relatively tight domestic monetary and fiscal conditions (including an appreciated real exchange rate) could potentially be having a restrictive effect on the domestic economy in terms of investments, labor inflows, external demand for goods and services, as well as the medium-term outlook for export-oriented sectors. This could contribute to more muted demand and a low inflationary environment persisting for longer.
Contractionary Supply • The potential for further escalation and worsening of the Middle East and Red Sea crises creates risk in terms of disrupting global supply chains (including oil extraction and shipping) that can feed through to higher global inflation. This could result in, on the one hand, a tighter-than-expected policy stance in advanced economies, and on the other hand, higher prices being transmitted to the domestic economy through the imported goods channel. • External trade-driven growth contributes to the sustained appreciation pressure on the dram. Any sharp reversal of such trends and slowdown in growth could generate inflationary risks stemming from the potential for significant financial outflows and increased volatility in financial markets.	Expansionary Supply • Growth in labor supply could bring the labor market back to equilibrium, generate excess supply, and contribute to declines in wages, meaningfully weakening inflationary pressures in the economy. This could be driven by a number of factors, including: ◦ a sharp reduction in Armenian migrant workers going abroad to Russia (given narrowing wage gaps between the two countries in key sectors, as well as the impacts of potential taxes on income earned abroad, which could push seasonal workers to participate in the Armenian labor market), ◦ the sharp acceleration in wages in recent years, the reduction in the role of remittances, as well as the structure of economic growth (which has enabled the absorption of workers of all types of skill levels into the labor market) could facilitate those outside of the labor force to quickly integrate into the labor force, thus easing inflationary pressures. • Deflationary impacts from seasonal food would persist, driven by growing aggregate output due to the rising share of greenhouses; if accompanied by potential difficulties in exporting certain agricultural goods, this could have further deflationary impacts on the economy and complicate the inflation management problem.

Box 4. Market Reference Scenario

Under the CBA Monetary Policy Framework (FPAS Mark II), the CBA does not prepare baseline projections. Instead, we rely on information from financial market participants (including commercial banks, fund managers, institutional investors, and so on)—compiled through quarterly surveys—to develop a macro-consistent “market reference scenario (MRS).” This MRS, in many ways, serves a similar purpose as a traditional baseline projection, in the sense that it represents the market’s rough estimation of the “most likely future,” given the information available to market participants at a particular point in time. The MRS can serve as a substitute for the baseline for those external stakeholders (e.g. IFIs, fiscal authorities, and others) who rely on exogenously-provided projections when making their own planning decisions. Through the use of the MRS, the CBA provides the opportunity to markets to not only understand third-party forecasts of the “most likely” future, but also pay attention to those types of risks, uncertainties, and scenarios that can pose risks from the perspective of monetary policy and price stability (which are presented in the preceding Taxonomy). At the same time, this provides an opportunity to effectively communicate the logic and rationale behind CBA’s policy decisions under the prudent risk management approach.

Note: The CBA Macro Survey is conducted on a quarterly basis by the CBA. For the purposes of informing the Market Reference Scenario, participants are each asked to provide a macro-consistent scenario, including a qualitative description of the narrative underlying the scenario and associated Case A- and B-type risks, as well as accompanying quantitative 12-quarter-ahead projections for CPI inflation, real GDP growth, the CBA policy rate, and the nominal AMD/USD exchange rate. The Q3 2024 survey was carried out in July-August 2024 and featured responses from 18 commercial banks, 35 credit organizations, 8 insurance companies, 19 investment companies, and 1 fund manager.

Figure B.4.1: CPI Inflation

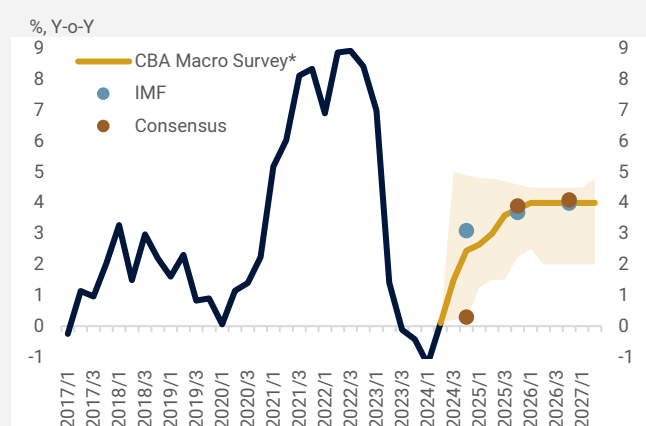


Figure B.4.3: Real GDP Growth Rate

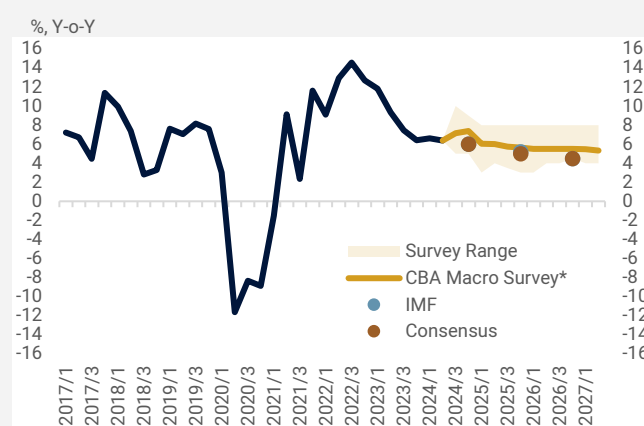


Figure B.4.2: CBA Policy Rate

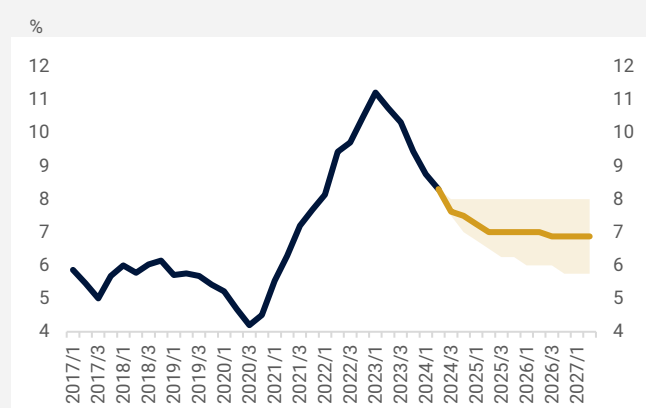
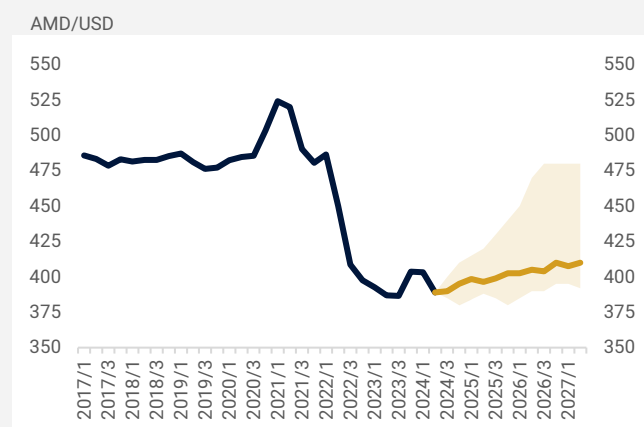


Figure B.4.4: Nominal AMD/USD Exchange Rate



* Note: CBA Macro Survey shown as average of survey responses (line) and range between min and max responses (shaded area).

Table B.4.1: Summary of Third-Party Projections

Source Date	IMF Apr-24	CBA Macro Survey Q3 24	Consensus Sep-24
GDP Growth (Annual Rate, %)			
2023 (Actual)		8.7	
2024	6.0	6.6	6.0
2025	5.2	5.9	5.0
2026	4.5	5.5	4.5
Inflation (Annual Rate, %)			
2023 (Actual)		-0.6	
2024	3.1	2.5	0.3
2025	3.7	3.8	3.8
2026	4.0	4.0	4.1

F. Monetary Policy Outlook

Case A: Interest Rates above Market Expectations

Uncertainties Regarding the Country Risk Premium and its Possible Upward Reappraisal

The illustrative Case A scenario is motivated by the uncertainties around the country risk premium estimates in Armenia, which remain at historical low levels mostly following emerging market trends. However the continuous regional tensions, uncertainties regarding the geopolitical developments as well as possibly weaker country fundamentals somewhat decoupled Armenia from the rest of the emerging market trends recently. In this context there is a possibility of quick adjustment of country risk premium to its long run sustainable level indicating a higher neutral interest rate in Armenia. In this scenario monetary policy should take into account the possibility of higher underlying riskiness and neutral rate in Armenia and adjust the policy stance. This will require a higher policy rate path compared to market expectations in order to avoid unnecessary volatility in financial markets due to differences in risk adjusted interest rates, as well as managing the risk of reversing the policy cycle in response to the possible adjustment of country risk premium to its long run sustainable value.

Emerging market bond yields and country risk premium estimates are typically influenced by monetary policy cycles and long-term interest rates in advanced economies. For instance, when there are expectations of tighter monetary policy in the U.S. and an increase in long-term interest rates, emerging market economies are often perceived as more risky during that period. This is because a strengthening U.S. dollar index tends to exacerbate public debt challenges in emerging markets. As the overall inflationary environment improves in the U.S. and risks to the full employment mandate begin to materialize, the likelihood of a "higher for longer" policy has significantly decreased. Consequently, market expectations for future interest rates, as well as long-term rates, have declined. This shift has made emerging market economies more attractive to financial flows, thereby reducing their country risk premium estimates. Despite this, the Congressional Budget Office's (CBO) projections of an exploding debt trajectory in the U.S. raise concerns about fiscal sustainability. This situation has made emerging market economies relatively more attractive in comparison, driving their country risk premiums lower than their long-term sustainable levels.

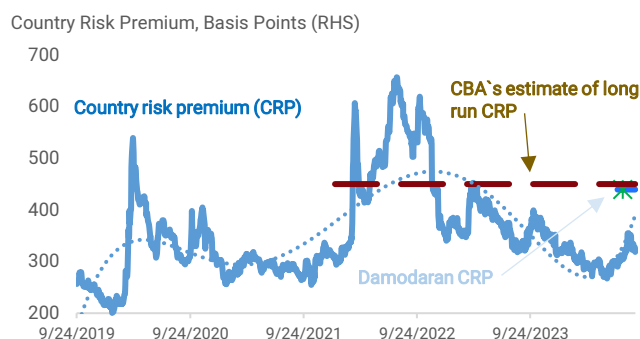
Armenia has generally tracked emerging market trends closely, with current risk premium estimates remaining at historically low levels. However, a combination of exogenous factors (such as ongoing regional tensions, uncertainties surrounding geopolitical developments in the region, Armenia's significant exposure to these potential changes, the uncertain outlook of the Russian economy, given Armenia's strong economic ties with Russia) and country-specific factors (including the structure of economic growth in Armenia, an uncertain growth outlook (see Box 2), and the risk of a higher public deficit over the medium term (see Section 3.B))—have recently caused a slight decoupling of Armenia from broader emerging market trends and contributed to a modest increase in the country's risk premium in recent weeks.

Due to the above mentioned regional and country specific factors Armenia may no longer benefit from positive external spillovers arising from lower interest rate expectations in advanced economies and redirected financial flows globally in favor of emerging markets

and there is a possibility of further decoupling of Armenia from this general positive trends. This type of scenario would imply a quick adjustment of Armenia's country risk premium from current low levels to more sustainable long run values (For example Damodaran's estimate of country risk premium for Armenia is 440 basis points, while CBA's estimates are around 450 basis points). If interest rates in Armenia don't follow the adjustment of country risk premium, (other factors being unchanged) this would imply that risk adjusted interest rates will be lower compared to foreign interest rates, thus creating arbitrage opportunities and triggering deeper financial outflows. In this scenario CBA should adopt a higher policy rate path compared to market expectations in order to account for possible upward adjustments in country risk premium and avoid unnecessary volatility in financial markets. Depending on how fast the adjustment will take place and how anchored are inflation expectations there are possible side effects arising from quick adjustment of country risk premium in terms of extra volatility in financial markets and possible de-anchoring of inflation expectations. While this is not the key narrative of Case A scenario, the CBA will react appropriately if such manifestations occur.

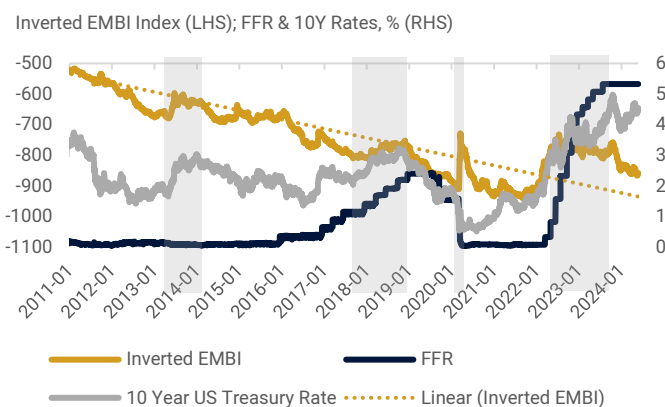
Figure 3.F.1:

Panel A. Estimated Country Risk Premium for Armenia and long-run sustainable level



Source: Bloomberg, CBA Estimates, Damodaran

Panel B. Yields for 10-Year US Treasury vs. EMBI



Source: FRED; Bloomberg

Case B: Interest Rates below Market Expectations

Weakening Labor Market Conditions and Negative Demand Environment in the Domestic Economy

Household survey results from 2024 and the slowing wage growth indicate a softening in labor market conditions in Armenia. The rise in the unemployment rate witnessed during the first quarter (see Figure 3.C.1) was not driven by a significant increase in labor supply, but rather by a decline in the number of employed individuals (see Figure 3.C.2), possibly being an early indicator of a weak demand environment coming to fruition. In the meantime, there has been significant moderation in private sector wage growth. Notably, excluding the financial and real estate sectors, wage growth has been negligible (see Figure 3.F.1).

Normally the labor market responds to economic developments with certain time lags, especially when it comes to weakening demand conditions, due to the presence of real and nominal rigidities, complicating the timely assessment of labor market conditions. Labor market data, obtained through household surveys, are published with significant delay, further complicating this effort. Therefore, there is considerable uncertainty around the real-time conditions of the labor market. On the other hand, the employment indicator, which is estimated based on the State Revenue Committee's databases and is growing at a high rate, only pertains to formal employment (based on registered employee numbers) and does not reflect the dynamics of the entire economy. What's more is that the aforementioned developments in this indicator can be a reflection of structural shifts in the labor market, where people are moving from informal (e.g., agriculture) to formal employment. This can also affect the average wage indicator, complicating the exercise of identifying and differentiating the impact of structural changes on private sector wages from the effects of other factors like demand, productivity, inflation, and inflationary expectations.

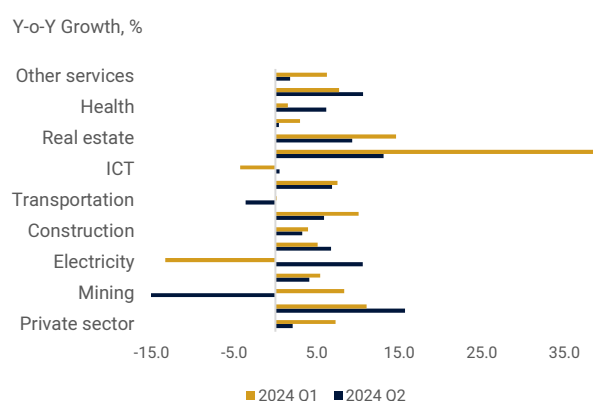
While not disregarding alternative perspectives and assessments of labor market conditions and taking into account significant uncertainties across other areas and the difficulties in assessing the environment around aggregate demand conditions (concentrated growth in specific industries, frequent revisions to growth projections, etc.), the Type B scenario is based in the assumption that the labor market is already exhibiting signs of an environment with gradual softening of demand. However, this weakening is masked by the high yet uneven (concentrated in few sectors) economic growth. In this scenario, a further increase in unemployment going forward and persistent low wage growth in a low-inflationary environment carry with them considerable uncertainty regarding domestic demand conditions, and bring about the risk of having an environment of deteriorating demand

(negative demand conditions) with sustained deflationary effects over the medium term.

The Case B-type scenario is predicated on the assumption that the current low-inflationary environment (see Figure 3.C.5) reflects the rather subdued domestic demand conditions and a further softening in the aforementioned labor market conditions could intensify deflationary pressures. Initial economic conditions under this scenario suggest that the output gap is in negative territory. While sustained weak demand over the forecast horizon is expected to have a limited impact on further downward price adjustments due to wage and price rigidities, persistently below-target inflation poses a risk of downward adjustments in inflation expectations, keeping them anchored at low levels, and further deepening the medium term restrained demand behavior beyond that horizon. Moreover, a decline in inflation expectations, all else being equal, would elevate real interest rates, potentially incentivizing more saving further restraining consumption.

In such a situation, to fulfill its mandate of achieving price stability while minimizing societal costs, the Central Bank must adopt a more aggressive and rapid reduction in the policy rate compared to current market expectations. Such action would aim to stimulate demand and bring inflation closer to the target over the medium term.

Figure 3.F.2: Growth Rate of Salaries in the Private Sector



Source: Statistical Committee of the Republic of Armenia, CBA

Summary of Illustrative Case A & Case B Macroeconomic Projections

Figure 3.F.3: Endogenous Interest Rate Path, %

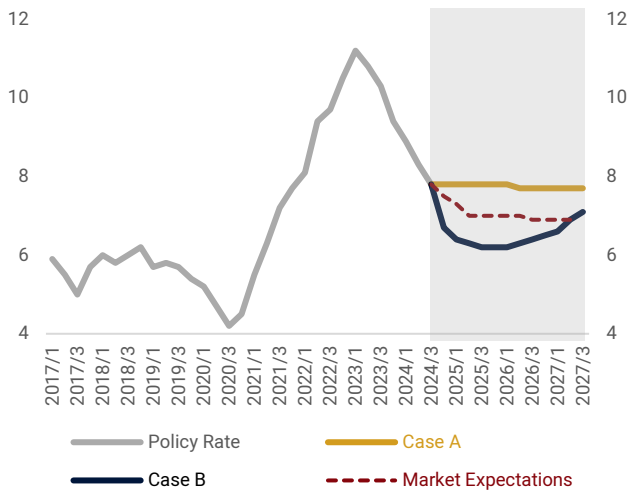


Figure 3.F.6: Non-Traded Sticky Price Inflation, Y-o-Y %

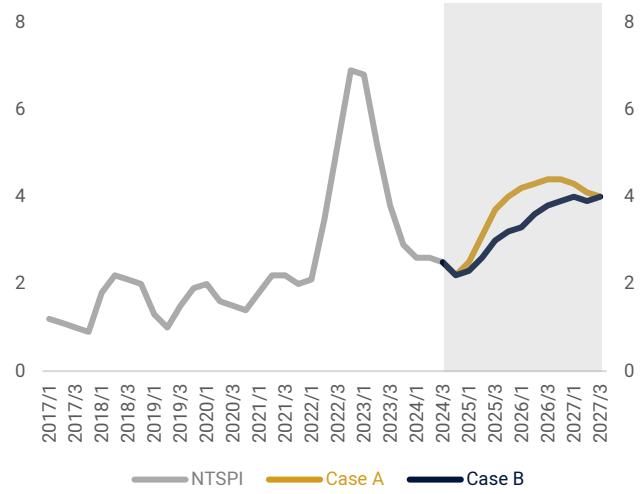


Figure 3.F.4: Real GDP Growth, Y-o-Y %

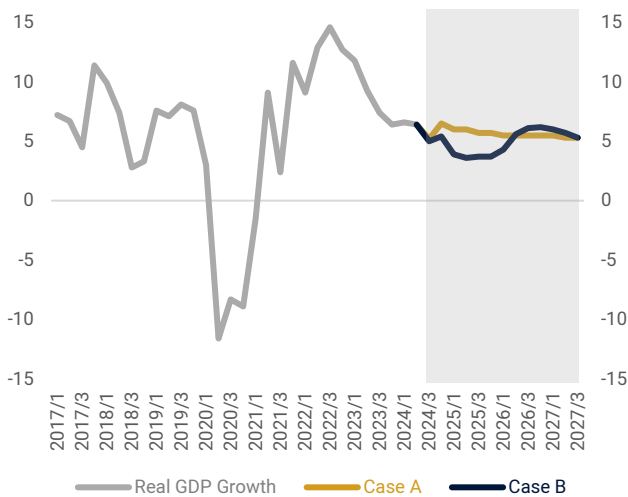


Figure 3.F.7: Output Gap, %

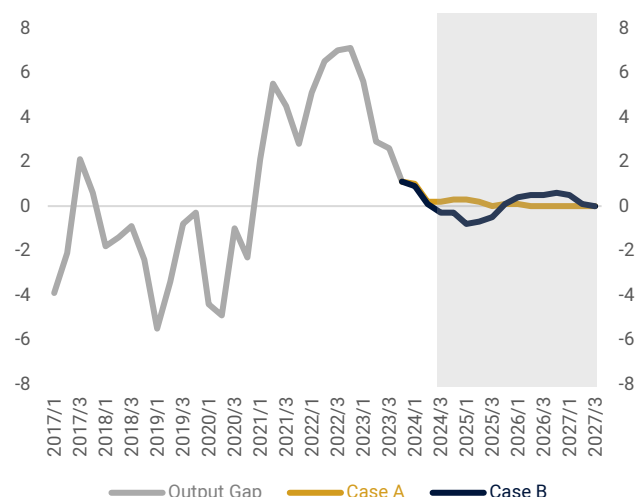


Figure 3.F.5: Headline CPI Inflation, Y-o-Y %

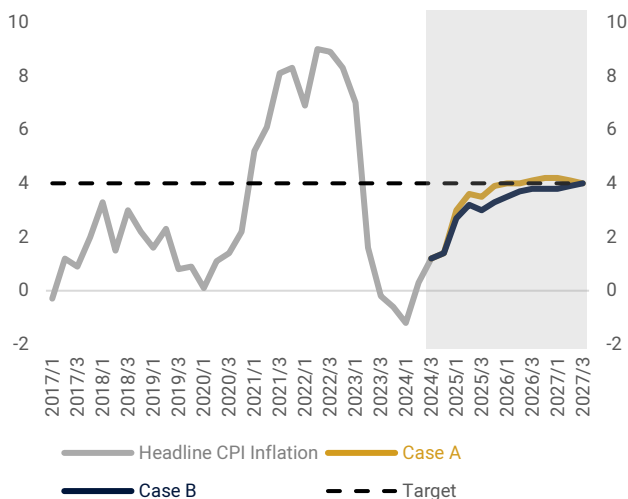


Figure 3.F.8: Real Interest Rate, %

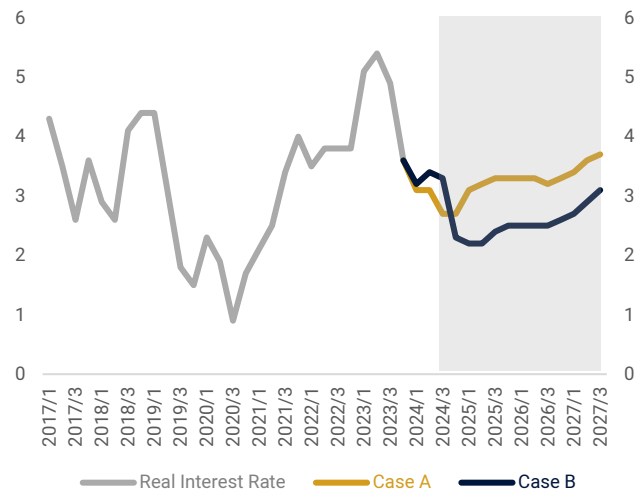


Figure 3.F.10: Nominal Effective Exchange Rate, Index

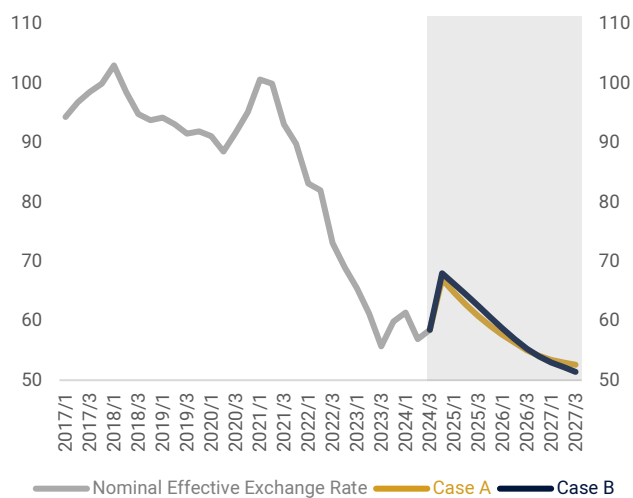


Figure 3.F.12: Model-Based 1-Year Ahead Inflation Expectations, Y-o-Y %

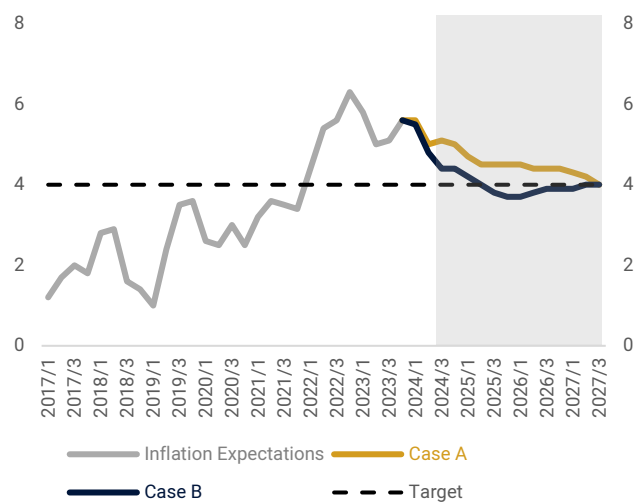


Figure 3.F.11: Real Effective Exchange Rate, Index

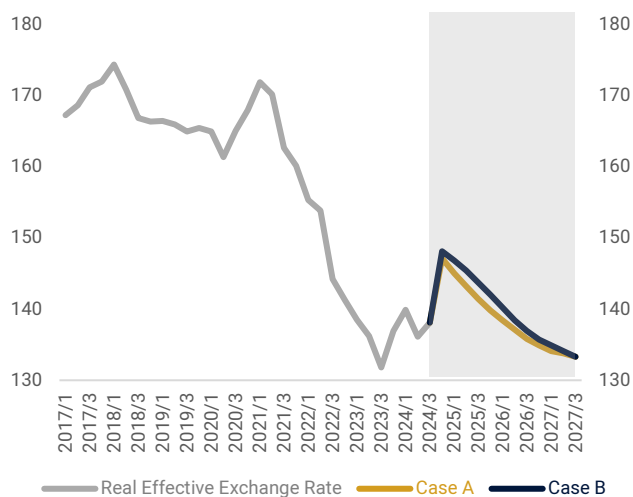


Table 3.F.1: Summary of Illustrative Macroeconomic Projections
Historical Data and Illustrative Case A & B Scenario Projections

		2022	2023	2024		2025		2026		2027	
		Actual	Actual	Case A	Case B	Case A	Case B	Case A	Case B	Case A	Case B
Global Economy											
USA economic growth	real growth, %	2	2.6	2.8	2.8	2.5	2.5	2.5	2.5	2.7	2.7
Eurozone economic growth	real growth, %	3.2	0.4	0.7	0.7	0.4	0.4	1.2	1.2	1.3	1.3
Russia economic growth	real growth, %	-2	3	4.1	4.1	1.7	1.7	1.4	1.4	0.9	0.9
USA inflation	average, %	7.7	4	3.0	3.0	2.7	2.7	2.3	2.3	2.0	2.0
EU inflation	average, %	8	5.4	2.6	2.6	2.4	2.4	2.1	2.1	1.9	1.9
Russia inflation	average, %	12.9	5.5	7.9	7.9	7.3	7.3	4.5	4.5	3.9	3.9
Oil price	dollar/barrel	99.5	82.6	84.6	84.6	85.6	85.6	86.5	86.5	87.8	87.8
Copper price	dollar/mt	8956	8472.1	9396.7	9396.7	10115.1	10115.1	10345.5	10345.5	10546.3	10546.3
FAO food price index	index	143.5	124.1	120.5	120.5	127.0	127.0	130.1	130.1	132.5	132.5
Domestic Economy											
Inflation											
Inflation	Y-o-Y, %, EOP	8.3	-0.6	2.1	2.0	4.0	3.2	4.2	3.8	4.0	4.0
CPI inflation	Y-o-Y, %, avg.	8.3	2	0.3	0.4	3.0	3.3	3.7	3.8	4.0	4.0
Non-traded sticky price inflation	Y-o-Y, %, avg.	4.5	4.8	2.5	2.5	3.3	2.8	4.3	3.7	4.0	4.0
Gross Income											
GDP	billion dram	8501.5	9453.2	10182.2	10120.5	11193.1	10874.9	12269.4	11932.0	13373.7	13125.2
GDP	real growth, %	12.6	8.3	6.5	5.8	5.9	3.7	5.5	5.5	5.4	5.5
Supply											
Industry	real growth, %	6.9	-2.2	4.7	4.5	5.6	3.5	4.9	5.1	4.9	4.9
Agriculture	real growth, %	-2.8	2.9	3.7	3.7	3.0	2.0	3.0	3.0	3.0	3.0
Construction	real growth, %	18.6	16.0	14.5	13.7	9.0	7.7	8.0	8.0	8.0	8.0
Services	real growth, %	18.4	11.5	6.7	6.5	6.2	3.5	5.5	5.0	5.5	5.5
Net taxes	real growth, %	8.1	11.0	1.1	0.6	6.3	3.5	5.5	5.5	5.2	5.2

Table continued on the following page.

Historical Data and Illustrative Case A & B Scenario Projections (continued from previous page)

		2022	2023	2024		2025		2026		2027	
		Actual	Actual	Case A	Case B	Case A	Case B	Case A	Case B	Case A	Case B
Demand											
Consumption	real growth, %	4.3	9.0	5.7	5.6	6.1	2.4	5.3	5.3	5.2	5.2
Public consumption	real growth, %	-2.2	28.3	-5.7	-5.7	8.6	5.6	5.0	5.0	5.0	5.0
Private consumption	real growth, %	5.6	5.6	7.9	7.2	5.8	2.0	5.3	4.8	5.3	5.3
Gross accum. of fixed assets*	real growth, %	14.0	10.1	9.1	9.1	1.6	-1.2	5.0	5.0	5.1	5.1
Public investment**	real growth, %	89.0	27.0	27.8	27.8	-8.7	-8.7	3.5	3.5	3.5	3.5
Gross accumulation of private fixed assets	real growth, %	-1.1	3.6	5.9	5.9	6.1	2.1	5.8	5.0	5.8	5.8
Export of goods and services	real growth, %	59.9	30.7	24.2	26.2	-31.5	-29.5	3.2	4.0	3.5	3.5
Import of goods and services	real growth, %	35.0	30.2	24.0	23.0	-28.4	-30.4	3.7	3.0	5.7	5.7
Current Account											
Balance of trade	million USD	-1859.2	-2340.6	-2393.4	-2263.4	-2776.0	-2336.0	-2873.5	-2405.9	-2972.1	-2475.1
Balance of services	million USD	1711	2167.9	1695.5	1695.5	1726.4	1886.4	1708.4	1704.3	1686.4	1492.8
Money transfers	million USD	1538.3	988.5	681.4	681.4	715.5	715.5	733.9	733.9	753.3	753.3
Current account	million USD	151	-476.7	-1026.3	-896.3	-1100.0	-500.0	-1197.2	-733.7	-1298.4	-995.1
Balance of trade	share of GDP, %	-0.8	-0.7	-2.7	-2.2	-3.8	-2.0	-3.9	-2.9	-4.0	-3.8
Balance of services	share of GDP, %	8.7	8.8	6.5	6.7	6.3	8.4	5.8	7.2	5.3	5.8
Money transfers	share of GDP, %	7.9	4	2.6	2.7	2.6	3.2	2.5	3.1	2.4	2.9
Current account	share of GDP, %	0.8	-1.9	-3.9	-3.5	-4.0	-2.2	-4.1	-3.1	-4.1	-3.9
Public sector***											
Revenues and grants	billion dram	2046	2358.6	2634.7	2634.7	2897.6	2897.6	3258.2	3258.2	3614.1	3614.1
Tax revenue	billion dram	1926	2221.9	2496.9	2496.9	2812.3	2812.3	3182.0	3182.0	3541.2	3541.2
Expenditures	billion dram	2243.5	2550.2	3117.6	3117.6	3513.4	3513.4	3812.1	3812.1	4086.8	4086.8
Deficit	billion dram	-197.5	-191.6	-482.9	-482.9	-615.8	-615.8	-553.9	-553.9	-472.7	-472.7
Revenues and grants	share of GDP, %	24.1	24.8	25.9	25.7	25.9	26.6	26.6	27.3	27.0	27.5
Tax revenue	share of GDP, %	22.7	23.4	24.5	24.3	25.1	25.9	25.9	26.7	26.5	27.0
Expenditures	share of GDP, %	26.4	26.8	30.6	30.4	31.4	32.3	31.1	31.9	30.6	31.1
Deficit	share of GDP, %	-2.3	-2	-4.7	-4.7	-5.5	-5.7	-4.5	-4.6	-3.5	-3.6

* The Central Bank only presents the indicator of gross fixed asset accumulation instead of gross accumulation, since the change in tangible working capital inventories is calculated by Armenia's Statistics Committee as a balancing item and it does not show the true level of gross accumulation. See https://www.armstat.am/file/article/sv_04_19a_112.pdf.

** Actual indicators of public investment are the capital expenditures of the consolidated budget, and the estimates are based on a revised macro-framework, 2025-2027, available at the time.

*** The 2024 budget indicators are the Central Bank of Armenia estimate. The 2025-2027 indicators are presented from the state Mid-Term Expenditures Framework.

